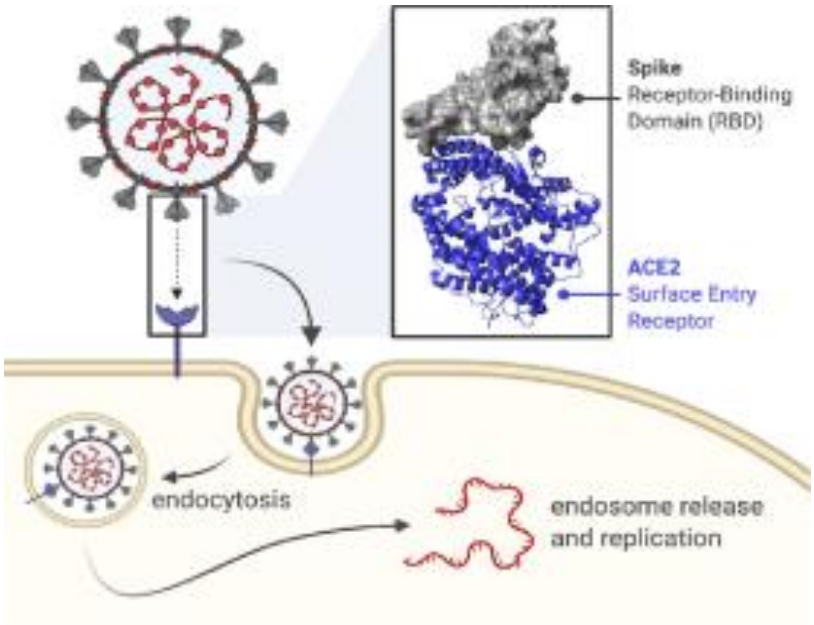
EXPLORING THE INTERACTION OF SARS-CoV-2 SPIKE GLYCOPROTEIN WITH HUMAN ACE2: AN OVERVIEW



SOUMIK ROY

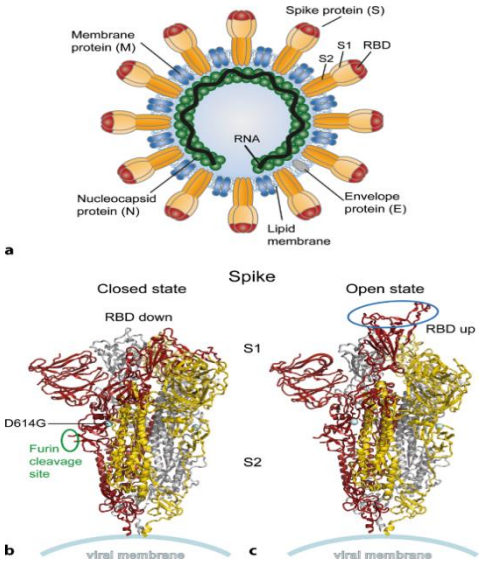
DEPARTMENT OF BIOTECHNOLOGY

ADAMAS UNIVERSITY

INTRDUCTION

The beginning of the outbreak at the end of 2019 witnessed a threat to public health and the world economy with coronavirus disease 2019 caused by severe acute respiratory syndrome coronavirus 2. By now, in June 2022, more than 500 million infections have been confirmed, with over 6 million deaths worldwide. [1,2,3] Other than the SARS-CoV-2, there are at least six other well-known CoVs that infect humans. These include: Human coronavirus NL63, HCoV-229E, HKU1, HCoV-OC43, SARS-CoV, and Middle East respiratory syndrome coronavirus. The first two coronaviruses, HCoV-OC43, as well as HCoV-229E, were discovered in the 1960s, whereas HCoV-NL63 and HKU1 are reported in the new past, in 2004 and 2005, respectively. These. However, with the emergence of SARS-CoV during 2002–2003, in which over 10% of all infected persons died, that picture changed dramatically [4,5]. About a decade later, MERS-CoV appeared with a high fatality rate of about 35% [6,7]. Now, nearly 20 years later, SARS re-emerged with SARS-CoV-2, which led to the current pandemic of COVID-19. Such highly pathogenic coronaviruses reflect the threat of emerging viruses in the 21st century.

The extreme acute respiratory syndrome, which was detected for the first time on approximately December 1st, 2019, in China, is the positive-sense single-stranded RNA virus, namely, the SARS-CoV-2 coronavirus [8,9]. Now, in a period of about six months, this betacoronavirus has spread over the globe, with over 14 million testing positive around the world, while the death toll surpasses 600,000 as of July 20th, 2020 (https://coronavirus.jhu.edu/map.html). The very close phylogenetic relationship of SARS-CoV-2 to SARS-CoV-1 is demonstrable, with almost 80% identicalness at the genomic level of the responsible severe acute respiratory syndrome outbreak beginning in 2002. Additionally, human SARS-CoV-2 at the whole genome level is >95% identical to a bat coronavirus, RaTG13, the natural reservoir host for multiple coronaviruses. Given the rapid emergence and dissemination of this pathogen, presently, there is no clinically approved, validated vaccine or SARS-CoV-2-specific targeting therapy although statins, heparin, and steroids show great promise to reduce fatality rates and antivirals probably shortening the duration of symptomatic disease presentation [10,11,12].



The SARS-CoV-2 uses, like most viruses, a trimeric Spike glycoprotein to affect the identification and binding to the host cell entry receptor and to mediate membrane fusion. Because of its role in directing the virus and facilitating entrance into the host

cell, the immunogen in most vaccines for the creation of neutralizing antibodies is the viral Spike protein, and thus this protein has the potential for the targeting of small molecules that would further inhibit host receptor binding and/or membrane fusion. Wild-type or enzymatically inactivated ACE2 has also been studied as a potential therapeutic biologic that may interfere with the infection cycle of ACE2-targeting coronaviruses. Therefore, a molecular understanding of SARS-CoV-2 Spike binding to ACE2 is critical for the description of mechanisms of viral binding and entry and to attempt the design of rational effective therapeutics. CoVs are enveloped, positive-sense, single-stranded RNA viruses infecting humans and animals.

They have the largest genomes among RNA viruses, ranging from 27 to 32 kb. There are four CoV genera: Alphacoronavirus, Betacoronavirus, Gammacoronavirus, and Deltacoronavirus. HCoV-229E and HCoV-NL63 belong to the Alphacoronavirus genus. The Betacoronavirus genus is divided into four subgenera: Embecovirus, Sarbecovirus, Merbecovirus, and Nobecovirus. HCoV-OC43 and HKU1 belong to Embecovirus; SARS-CoV and SARS-CoV-2 to. The four main structural proteins are spike, envelope, membrane, and nucleocapsid. The N protein is related to the viral genome RNA and ensures stability. The E and M proteins develop part of the viral envelope and participate in virus assembly. The S glycoprotein extends out from the viral envelope, interacting with receptors on host cells and mediating viral cell membrane fusion to permit entry.

Consequently, it is of prime importance for CoVs in virulence, tissue tropism, and host range. It is also one of the major targets for the host immune response, and thus becomes a focus for vaccine and therapeutic development. The Spike glycoprotein of SARS-CoV-2 is a heterodimer of two subunits: a receptor binding subunit, S1, and a membrane fusion subunit, S2.

The Spike glycoprotein forms stable homotrimers, each consisting of 66 canonical sequons for N-linked glycosylation and a variable number of potential O-linked glycosylation sites. Quite interestingly, coronaviruses virions bud into the lumen of the endoplasmic reticulum-Golgi intermediate compartment, ERGIC, raising unanswered question regarding the precise mechanisms that glycoproteins on the viral surface are processed as they make their way through the secretory pathway (13). Although this and related studies are approached logically with the aid of recombinant proteins, one previous study of SARS-CoV-1 revealed that the intracellular budding can affect glycosylation of the Spike and has never been investigated in SARS-CoV-2. This report summarizes crucial research elucidating the structural basis of S-mediated cell entry for SARS-CoV-2.

It discusses aspects of the evolution of the S glycoprotein for the understanding of CoV cross-species transmission from a structural biology perspective. Furthermore, it recently summarizes the work on the drug therapeutics against the SARS-CoV-2 S glycoprotein, including

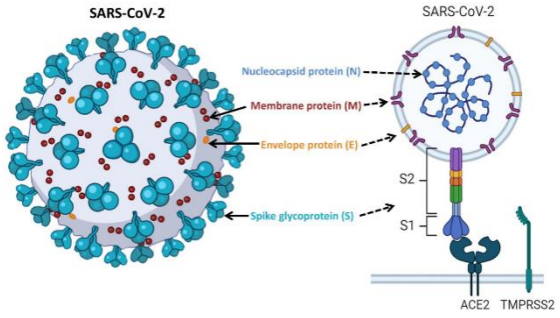
antibodies and nanobodies, among other nanopeptide inhibitors. A more detailed analysis of the binding of the SARS-CoV2 Spike protein to ACE2 will be pivotal for the effective conception of therapeutic interventions and for the design of vaccines against this and future coronavirus pandemics.

Role of Spike Glycoprotein in SARS-CoV-2

The spike glycoprotein plays a very central role in SARS-CoV-2 infections. Considered one of the main components of the ability to cause COVID-19 by the SARS-CoV-2 virus, it is termed a primary target for the development of therapeutics and vaccines. This surface protein plays a central role in the process of infection and is, therefore, key to understanding the pathogenicity of the virus and consequently how to fight it.[13]

Structure and Function

The Spike protein is a large transmembrane protein protruding from the virus surface, responsible for the crown appearance of the virus. It is about 180 kDa in size and exists as a trimeric form composed of three identical subunits. Each S-protein monomer is made up of two major domains: the S1 and S2 subunits.



The S1 subunit contains a receptor-binding domain that specifically interacts with the ACE2 receptor on human cells, while the S2 subunit mediates membrane fusion of the host and viral membranes to enable the entry of the latter. This interaction is a very key step to have infection take place in host cells.

Mechanism of Viral Entry

The entry process of SARS-CoV-2 into host cells involves various key steps:

1. Attachment: The RBD in the S1 subunit binds with high affinity to the ACE2 receptor on the surface of the host cells.

2. Conformational Changes: Binding induces a conformational change that exposes the S2 subunit.

3. Fusion: The exposed S2 subunit mediates the fusion of the viral envelope with the host cell membrane, allowing the release of viral RNA into the host cell's cytoplasm.

4. Replication: The viral RNA hijacks the host cell machinery for replication and the synthesis of new particles.

Role in Pathogenicity

The pathogenicity of SARS-CoV-2 is central to its interaction between the Spike glycoprotein and ACE2. Especially the S-protein variations, such as mutations in the RBD of the protein, may alter its binding affinity for ACE2 and therefore change its infectivity. For instance, E484K and N501Y have been associated with increased transmissibility.

It is also the S-protein that triggers the immune response of the host, hence a major target for vaccines and therapeutic antibodies. Indeed, non-neutralizing and neutralizing antibodies against the S-protein have already been shown to neutralize the virus by inhibiting its interaction with ACE2.

Vaccine and Therapeutic Target

COVID-19 vaccines, much like those from Pfizer-BioNTech and Moderna, induce an immune response to Spike glycoprotein. The vaccine takes a form of the S protein to induce immunity but not cause a disease. It is targeted by therapeutic interventions, such as monoclonal antibodies, to prevent entrance into the cells and reduce the severity of the disease.

Role of Human ACE2 Receptor

The human ACE2 receptor is critical for the SARS-CoV-2 infection that leads to COVID-19. Knowing its function and interactions may give an idea of how the virus gets inside human host cells, thus leading to progress in causing a disease.

Structure and Function

ACE2 is a transmembrane enzyme involved in the renin-angiotensin system, controlling blood pressure and fluid balance. ACE2 is a single-pass transmembrane protein with an extracellular domain that contains the catalytic site, interacting with various physiological substrates. One of these substrates is angiotensin II, a peptide involved in blood pressure regulation.

Apart from the enzymatic role, ACE2 acts as the major receptor for SARS-CoV-2. This attachment of the receptor binding domain of the viral Spike glycoprotein, otherwise known as S-protein, to ACE2 happens in a selective and high-affinity manner. This interaction is very essential for the entry of the virus into host cells.

Mechanism of Viral Entry

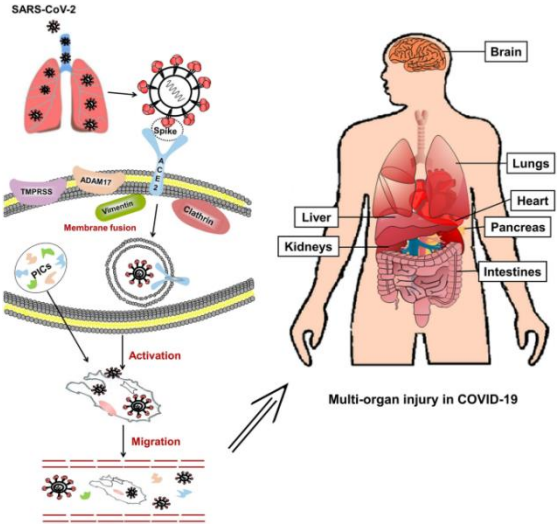
Viral entry of SARS-CoV-2 into human cells happens in several mechanistic steps:

1. Binding: Binding of the RBD of S-protein to the extracellular domain of the ACE2 receptor on host cells is a very specific interaction that tightly attaches the virus to the host cell membrane. [14]

2. Conformational Changes: A conformational change in the S-protein exposes a fusion peptide, and this modification in the protein is very significant for the remaining steps in viral entry.

3. Fusion: The exposed fusion peptide enables the fusion of the viral envelope with the cell membrane of the host cell. This fusion allows the viral RNA to enter into the host cell and begin replicating itself, forming new particles.

4. Internalization: Following fusion, there is internalization of the virus inside the host cell, after which there is a release of viral RNA followed by the commencement of the cycle of the phase of viral replication.



Role in Pathogenesis

ACE2 is highly expressed in respiratory tract epithelia, heart, kidneys, and intestines in the human body. This broad expression contributes to the multifarious manifestations seen in COVID-19, which includes not only affection of the respiratory system but also many other organs. The interaction between the Spike protein and ACE2 can result in inhibition or reduction of ACE2 function, leading to an imbalance of the renin-angiotensin system. Disruption of ACE2 function might further drive the inflammation and tissue injury, among other mechanisms of pathology underlying severe COVID-19.

It is, however, also indicated that the expression level of ACE2 alone modulates the severity of infection. The high expression levels of ACE2 in some tissues might further facilitate higher degrees of viral entry, leading to the development of more serious diseases. In contrast, low expression could be associated with reduced susceptibility to infection. [15]

Treatment and Prevention Implications

Such knowledge on ACE2 in the course of SARS-CoV-2 infection bears crucial implications for both treatment and prevention strategies. Therapies that prevent the binding of the S-protein to ACE2, including monoclonal antibodies to Spike protein, may interfere with the entry of the virus into the host cell. Additionally, therapy development is underway with ACE2 modulators and inhibitors aiming to reduce the after-effects of viral infection and to restore the balance of the renin-angiotensin system.

It is on this knowledge that these vaccines also take advantage of in targeting the Spike protein, inducing an immune response that will prevent binding to ACE2. Targeting the S-protein makes vaccines protective from the virus and reduces the chances of severe diseases.

OBJECTIVE OF STUDY

AIM AND OBJECTIVE

Aim:

This project will look into the molecular level of the interaction between the spike glycoprotein of SARS-CoV-2 and the human ACE2 receptor.

Objectives:

1. Structure-Function Analysis: Computationally analyze the structural feature of the spike protein of SARS-CoV-2, along with analysis of its interaction with the ACE2 receptor, using bioinformatic tools.

2. Bioinformatics Application: Apply and assess various computational tools to study these molecular interactions, contributing to the broader understanding of the virus's mechanism of action.

TOOLS USED

[A]. Sequence and Structural Data Resources

Sequence and Structural Data Resources Findings refer to the insights and data obtained from databases like NCBI, RCSB PDB, and UniProtKB. Detailed sequences of SARS-CoV-2 Spike glycoprotein and Human ACE2 are included from databases such as NCBI, RCSB PDB, and UniProtKB, outlining 3D structural models, functional annotations, and molecular interaction between these two proteins. The data shows precisely how the two proteins interact at a molecular level, pinpoints exact key sites on these proteins involved in the binding, and gives information about their biological roles and evolutionary relationships. This information is very critical to the understanding of mechanisms of entry, design of therapeutic interventions, and may provide potential drug targets.

1. National Center for Biotechnology Information (NCBI):

The NCBI has become one of the most critical resources that provides access to almost all types of biological data, including genetic sequences, protein structures, and functional annotations. NCBI, through its rich databases like GenBank and the Protein Data Bank, provides broad information on sequences for different organisms, including SARS-CoV-2 and Human ACE2. GenBank provides annotated DNA and RNA sequences, while PDB contains detailed three-dimensional structures of proteins and nucleic acids, thereby defining, at the molecular level, interactions of the Spike glycoprotein with ACE2. NCBI tools put options for sequence alignment, functional annotation, and comparative genomics at

one's disposal. These enable a whole diversity of bioinformatics analyses.

2. RCSB PDB:

The Research Collaboratory for Structural Bioinformatics Protein Data Bank (RCSB PDB) is a repository for three-dimensional structural information on biological macromolecules. It offers atomic-resolution structures, resulting from X-ray crystallography, NMR spectroscopy, and cryo-electron microscopy. Hence, associated with SARS-CoV-2 Spike glycoprotein and Human ACE2, RCSB PDB can give high-resolution structures that explain the spatial arrangement of these proteins. Atomic details of protein-protein interactions, recognition of binding sites, or investigations on conformation changes can be studied by researchers. The RCSB PDB enables a variety of visualization and structure analysis tools so that one can examine how structure data supports function and interaction implications.

3.UniProtKB: UniProtKB (Universal Protein Knowledgebase) is a comprehensive resource for protein sequences and functional information. For interaction analysis between SARS-CoV-2 spike protein (S-protein) and human ACE2, UniProtKB provides detailed annotations on protein structure, domains, and post-translational modifications. It offers curated data on the SARS-CoV-2 S-protein (UniProt ID: P0DTC2) and human ACE2 (UniProt ID: Q9BYF1), facilitating the exploration of binding regions and structural motifs. Additionally, I accessed the PDBsum database through UniProt to generate the Ramachandran plot, aiding in the evaluation of protein structure and backbone conformations.

[B]. Domain and motif analysis is important to understand the functional and structural properties of proteins. They identify regions in proteins critical to their biological activity or interactions. In particular, this analysis becomes very important in exploring the interaction between the Spike glycoprotein of SARS-CoV-2 and Human ACE2, since it identifies key regions that enable binding and function.

1. Conserved Domains Database and Resources:

Conserved Domains Database is a resourceful tool in attempts at identifying and mapping protein domains. Domains are the independent structural and functional units of a protein that, sometimes, could be linked to particular biological functions. Such tools as CDD identify conserved domains in protein sequences using position-specific scoring matrices. Analysis of SARS-CoV-2 Spike glycoprotein and Human ACE2 using CDD identified domains responsible for viral binding and entry. It could, therefore, be such domains that contribute to the evolutionary conservation of protein functions between different species. This would be important in understanding how the virus interacts with the host cell

receptors and may present potential targets for drug development.

2. MOTIF: Searching Protein Sequence Motifs:

MOTIF is a tool for the identification of conserved sequence motifs within proteins.



Those are short sequences repeated within a protein, most often critical for the protein's functions, stability, or even interaction with other molecules. Among others, it can be applied to the Spike glycoprotein or ACE2, showing motifs concerned with receptor binding or other important functionalities. These motifs are very well conserved among different strains and even between species, pointing out the necessity of their maintaining fundamental protein functions. It can also be used in identifying such motifs that will further help in designing drugs and vaccines targeted against certain regions essential for viral entry or replication.

[C]. Sequence Alignment and Prediction

Sequence alignment and prediction are basically carried out to portray evolutionary relationships, structural conservation, and functional regions of proteins. They enable the comparison of protein sequences to identify conserved regions and, from these, to infer structural and functional features. This is especially important in cases like that of SARS-CoV-2 Spike glycoprotein and Human ACE2, whose similarities in sequences might give insights into their interaction and evolutionary history.

1. T-Coffee Server:

T-Coffee stands for Tree-based Consistency Objective Function for



alignment Evaluation. It is a full-featured, very well-versed sequence alignment tool that can produce very accurate alignments by combine results from several methods. T-Coffee is especially useful for aligning sequences that are distantly related or have low similarity, which is often the case with fast-evolving viruses like SARS-CoV-2. Using T-Coffee to align the Spike glycoprotein and ACE2 sequences will return evolutionary pressures acting on these proteins and identify critical residues involved in their interaction.

- **PHYLOGENETIC ANALYSIS (MEGA XI)**



Sequence Alignment and Phylogenetic Tree Construction

In my phylogenetic study, I used MEGA to perform alignments of the Spike protein sequences from various coronavirus strains, like SARS-CoV, MERS-CoV, and SARS-CoV-2. This was a key process in finding out which areas are conserved and which mutations decrease or increase the binding affinity with the human ACE2 receptor. By doing the alignments of the

sequences, I managed to identify those amino acids that have been preserved among the different strains and are therefore possibly responsible for the integrity of the structure and functionality of the Spike protein.

Maximum Likelihood Method and Bootstrap Analysis

The phylogenetic tree was generated using the Maximum Likelihood method. This relationships is displayed as a bootstrap value associated with each branch; it is a quantification of the probability that a given branch exists.

NCBI BLAST and T-Coffee

I used BLAST to retrieve homologous Spike protein sequences from various coronavirus species to ensure a dataset that is diverse enough for phylogenetic analysis. This tool is efficient in finding proteins that belong to the same family or that are closely related from different species. Afterwards, I aligned them using T-Coffee, which provided minute alignment by highlighting conserved domains and variable regions that enhanced the analysis of similarities and differences in sequences.

3. EMBOSS Explorer:

EMBOSS Explorer is a power package of tools for the sequence analysis of proteins and nucleic acids. It incorporates many facilities, including alignment, motif discovery, and much more. Its interface is very easy to use in running operations like

statistical tool is very strong in inferring evolutionary relationships from sequence data. The ML approach will now provide the estimate of phylogenetic trees by finding a tree topology that maximizes the likelihood of observed sequence data under a given model of evolution. I performed bootstrap analysis, where the test is replicated on the reshuffled original data to obtain a distribution of tree topologies. The confidence placed in the inferred

sequence alignment, translation, or any other analyses.

Tools offered by EMBOSS

1. *EMBOSS Water*: Does a local pairwise alignment to highlight regions in two protein sequences of quite some similarity.

2. *EMBOSS Dotmatcher*: It produces a dot plot comparing the graphically identical regions of two sequences.

3. *EMBOSS Pepstats*: This will check the physicochemical properties of a protein against the interacting potentials.

4. *EMBOSS Antigenic*: Predicts the potential antigenic regions on a protein that may overlap with the interaction site

[E]. Antibody and Epitope Prediction

Immunoinformatics also contains Antibody and Epitope Prediction tools to identify the regions that antibodies and T-cells identify in antigens. This involves the immune responses, and thus, it is particularly important for the design of vaccines and therapeutic antibodies for a wide range of infectious diseases, whereby COVID-19, the targeting of the SARS-CoV-2 Spike

glycoprotein, has to be very specifically targeted.

1. - IEDB Analysis Resources (B Cell Tools and T Cell Tools):

IEDB Analysis Resources provide a comprehensive set of tools for the prediction and analysis of immune epitopes. B Cell Tools focus on predicting linear and conformational B cell epitopes, which are critical for antibody binding and immune response. These tools help identify regions of proteins likely to be recognized by B cells. T Cell Tools, on the other hand, predict peptides that can be presented by MHC molecules to T cells, crucial for adaptive immunity.

- **IgPred Tool:** IgPred is a bioinformatics tool developed to predict potential and specific antibody-specific B-cell epitopes.



In the context of your project dealing with interaction between SARS-CoV-2 Spike glycoprotein and human ACE2, using IgPred may enable better insights into potential antibody binding sites along the Spike protein through scanning of the protein sequence as this tool predicts the likelihood of B-cell epitope formation important for vaccine design and the development of therapeutic antibodies. This tool exploits machine learning algorithms that incorporate the features of the sequence, and it can be an extremely precious resource in identifying regions that may provoke immune responses to the virus, which in turn will favor the fight against COVID-19.

METHODOLOGY AND RESULT

[A]. In the study of protein interactions, particularly in the context of viral proteins and their human receptors, it is essential to utilize various computational methods to understand the underlying mechanisms. The interaction between the SARS-CoV-2 Spike glycoprotein (S-protein) and the Human ACE2 receptor is a critical focus in virology and immunology. This section explores different computational techniques used to analyze and visualize the similarities and interactions between these proteins.

SEQUENCE ALIGNMENT AND SIMILARITY SEARCHES

Sequence alignment is a fundamental technique for comparing proteins and understanding their evolutionary relationships. Methods for sequence alignment, such as those developed by Needleman and Wunsch (1970), utilize dynamic programming to find optimal alignments between sequences. This approach involves creating a scoring matrix where each cell represents the alignment score between prefixes of the two sequences.[18]

```
--GTGACATGCGAT--AAGAGG---CCTT--AGATCCGGATCTT
  | ||| ||||||  |||      ||||  ||||| |  ||||
GGGAGAC-TGCGATACAAG---TTACCTTGTAGATCTG-TCTT

                Key: - gap
                  | match
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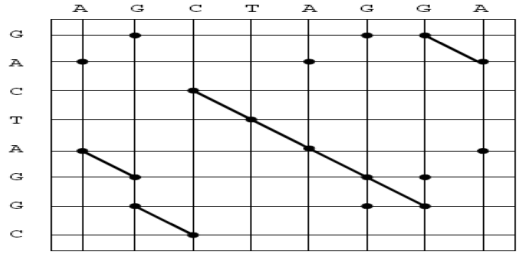
Figure 1: Example of sequence alignment

For analyzing the Spike glycoprotein and ACE2 receptor, sequence alignment can reveal conserved regions and potential interaction sites.

The Needleman-Wunsch algorithm, employed by tools such as NCBI BLAST, was used to compute the conserved regions and interaction sites between the SARS-CoV-2 Spike protein and the ACE2 receptor. PAM and BLOSUM matrices provided quantification of amino acid substitutions, aiding in understanding functional and structural significance.

DOT-MATRIX ANALYSIS

Dot-matrix plots are useful for visualizing regions of similarity between two sequences. By plotting a dot for every match between the sequences along a matrix, one can visually identify regions of high similarity or repeated motifs. [19]



Using EMBOSS Dotmatcher, graphical dot plots were generated to visualize repeated motifs and sequence similarities, revealing potential binding sites that are key to the interaction between the spike glycoprotein and ACE2.

DYNAMIC PROGRAMMING ALGORITHM

Dynamic programming algorithms, such as the Needleman-Wunsch algorithm, can be employed to align sequences of the Spike glycoprotein and ACE2 receptor. This technique computes an optimal alignment by breaking the problem into simpler subproblems, leveraging previously

computed alignments to build up the solution for longer sequences. The algorithm uses a scoring system for matches, mismatches, and gaps to evaluate the best possible alignment.[20]

SCORING MATRICES AND GAP PENALTIES

In protein sequence alignments, scoring matrices like PAM and BLOSUM are essential for assessing the biological significance of alignments. PAM matrices provide evolutionary-based scores for amino acid substitutions, which are useful for understanding how mutations may affect protein function.

Tools like EMBOSS Pepstats and EMBOSS Antigenic utilize substitution matrices, such as PAM250, to evaluate the evolutionary significance of amino acid changes, offering insights into mutations that may affect the binding affinity and interaction dynamics of the proteins.[21]

PAM MATRICES AND LOG ODDS SCORES

PAM matrices, such as PAM250, are used to score amino acid substitutions based on evolutionary data. For analyzing the interaction between SARS-CoV-2 Spike and ACE2, PAM matrices help determine the likelihood of specific amino acid changes and their potential impact on protein interactions. Log odds scores, derived from PAM matrices, offer a convenient way to compare the probability of amino acid substitutions, aiding in the identification of functionally relevant regions in the proteins. [22]

STRUCTURAL AND PHYLOGENETIC ANALYSIS:

Structural analysis was performed using Chimera for in-depth visualization of the spike glycoprotein and ACE2 interactions, aided by homology models generated through SWISS-MODEL. Additionally, phylogenetic relationships were explored through the construction of a phylogenetic tree using tools like MEGA 11, providing a comparative view of the evolutionary trajectories of these proteins.

In summary, the application of sequence alignment methods, dot-matrix analysis, dynamic programming algorithms, and scoring matrices provides a comprehensive approach to studying protein interactions. For the SARS-CoV-2 Spike glycoprotein and Human ACE2 receptor, these computational tools are instrumental in elucidating the mechanisms of binding and interaction, ultimately contributing to a deeper understanding of viral entry and potential therapeutic targets.

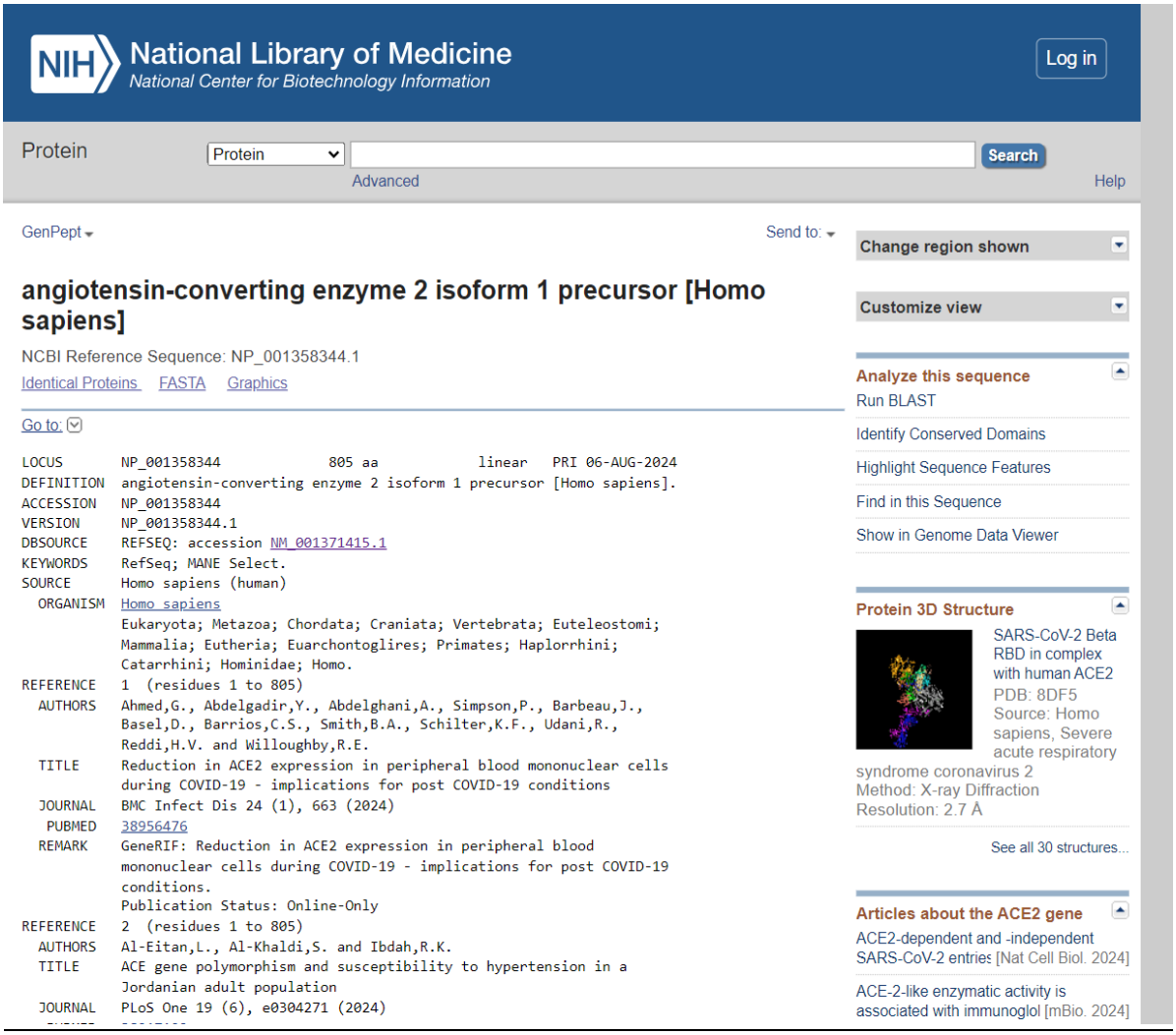
[B]. BIOINFORMATIC TOOL ANALYSIS

(i). Sequence and Structural Data Resources

- The amino acid sequences of the SARS-CoV-2 Spike glycoprotein (*AC. No. - P0DTC2.1*) and human ACE2(*AC. No. – NP_001358344.1*) were retrieved from **National Center for Biotechnology Information (NCBI)**: Sequence data and functional annotations were accessed using the NCBI database for additional insights into the proteins.

- Search for protein sequences at the NCBI website:
- Spike: P0DTC2.1 and ACE2: NP_001358344.1

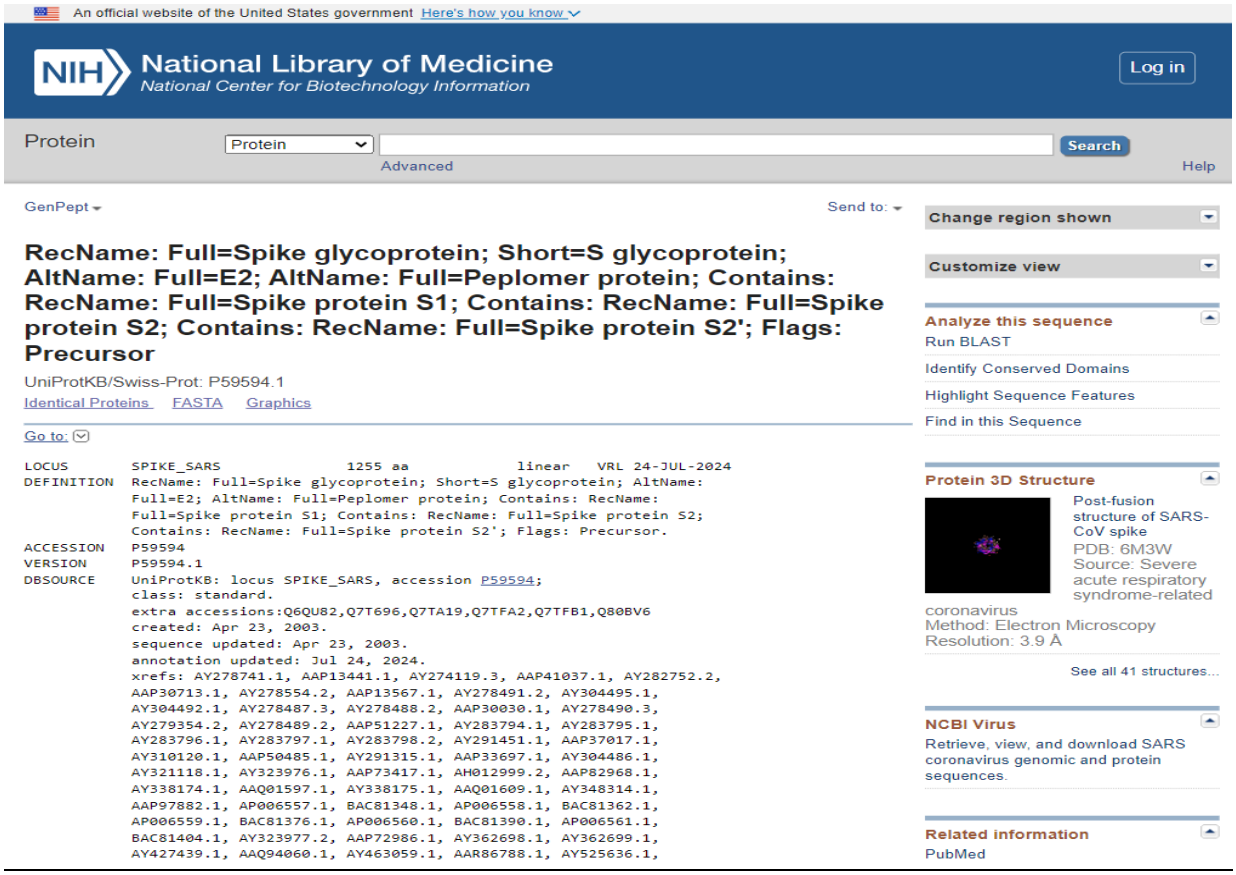
- Copy and Paste the FASTA formats in text format in notepad and save it to use for later analysis.
- Functional annotation of the proteins provides the necessary background on structural domains and interaction.



FASTA Sequence: -

>NP_001358344.1 angiotensin-converting enzyme 2 isoform 1 precursor [Homo sapiens]

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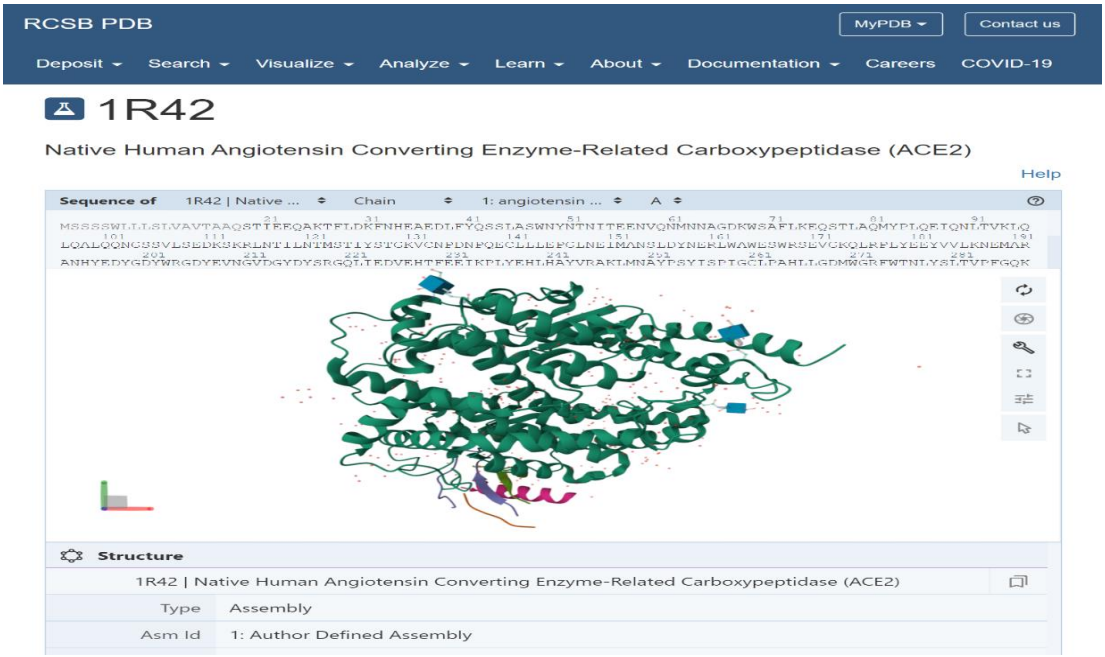
>sp|P0DTC2.1|SPIKE_SARS2 RecName: Full=Spike glycoprotein; Short=S glycoprotein; AltName: Full=E2; AltName: Full=Peplomer protein; Contains: RecName: Full=Spike protein S1; Contains: RecName: Full=Spike protein S2; Contains: RecName: Full=Spike protein S2'; Flags: Precursor

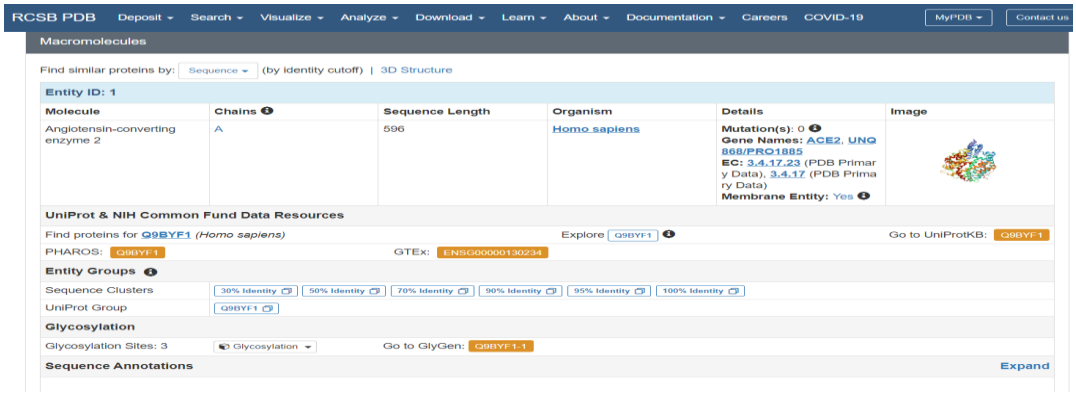
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TSCCSCLKGCCSCGSCCKFDEDDSEPVLKGVKLHYT

- **RCSB PDB:** RCSB PDB was used for retrieving protein structure and functional information. For structural analyses, 3D structures of these proteins were accessed from the Protein Data Bank (PDB) using entries 7F3Q (SARS CoV-2 Spike protein S2) and 1R42 (Human ACE2).

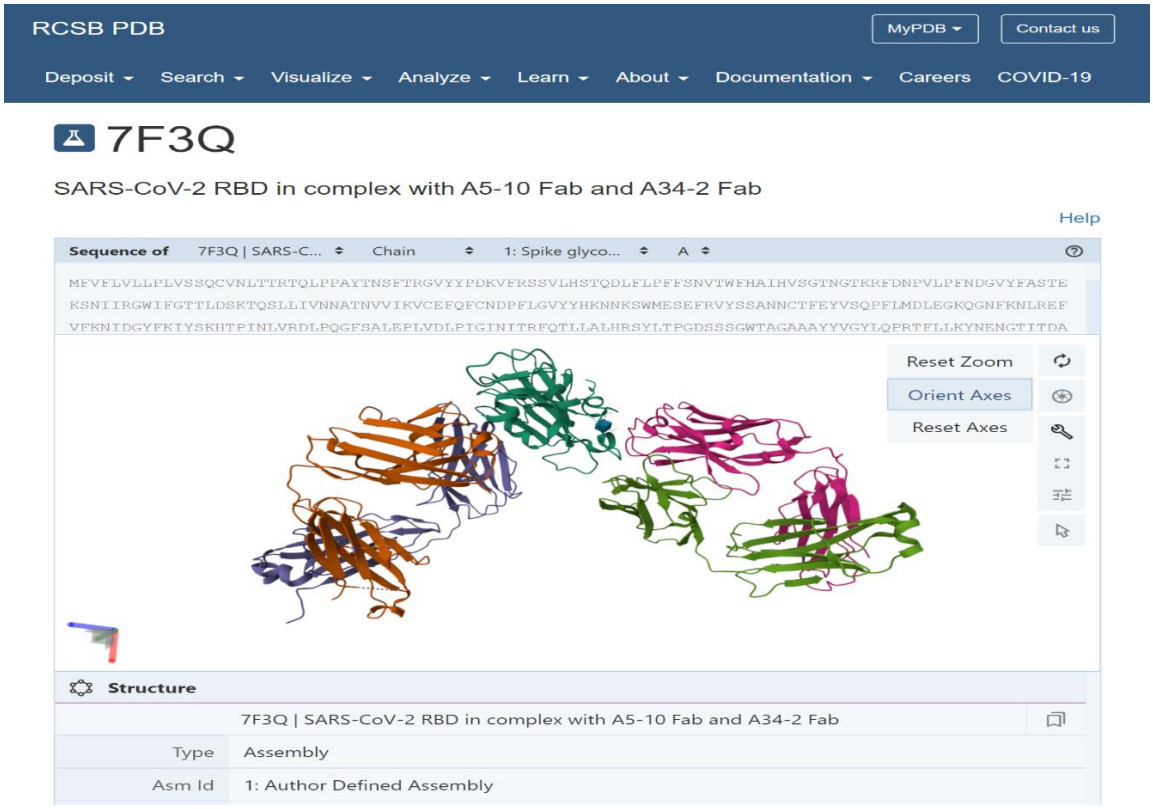
    - Go to RCSB PDB website
    - search by PDB IDs 7F3Q for Spike S2 and 1R42 for ACE2.
    - Download the 3D structures and do the structural visualization and functional site analysis with them in the site itself.

Ribbon Structure of Human ACE2 Protein Receptor and its analysis section from RCSB PDB

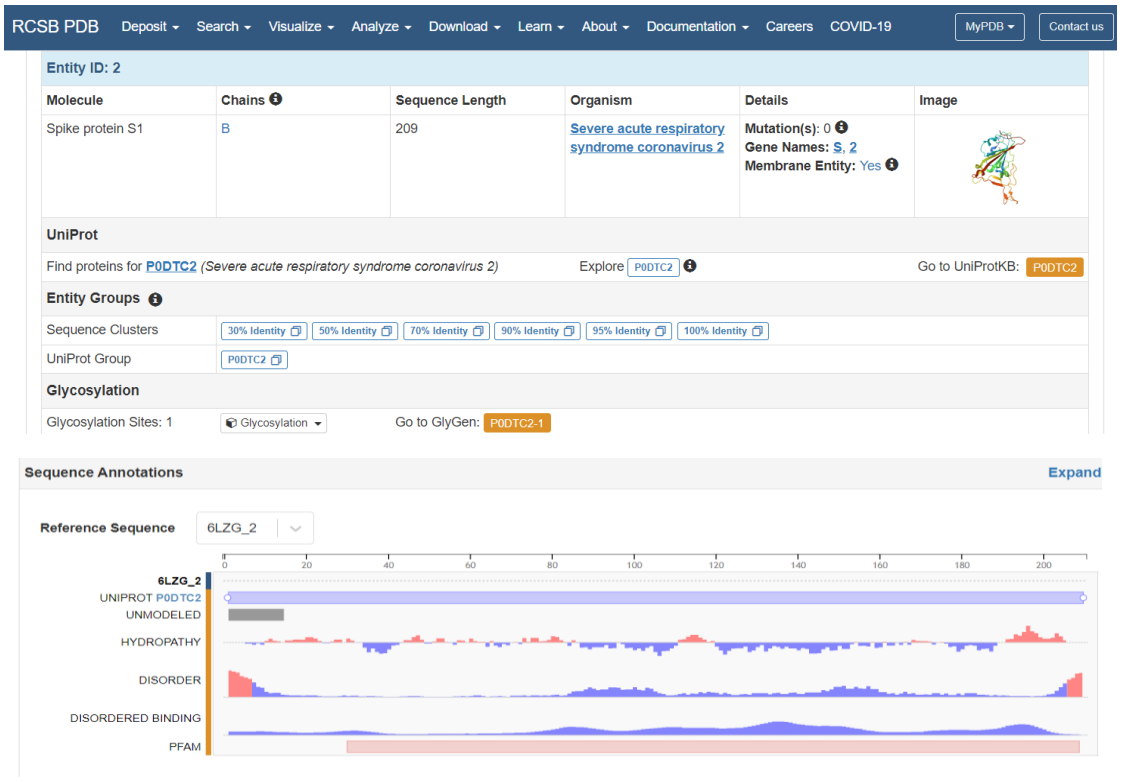




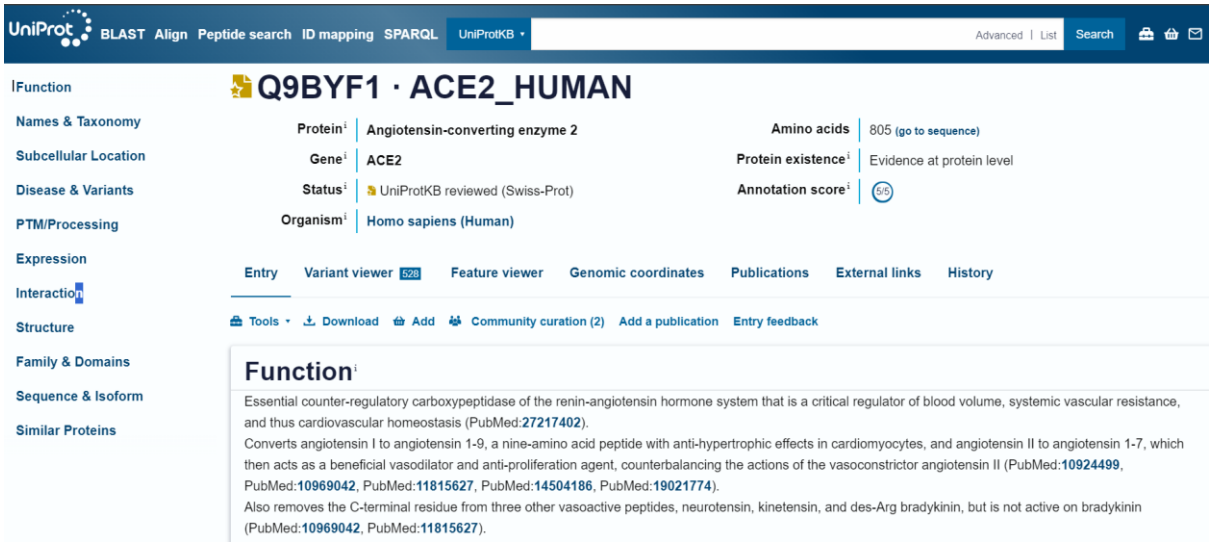


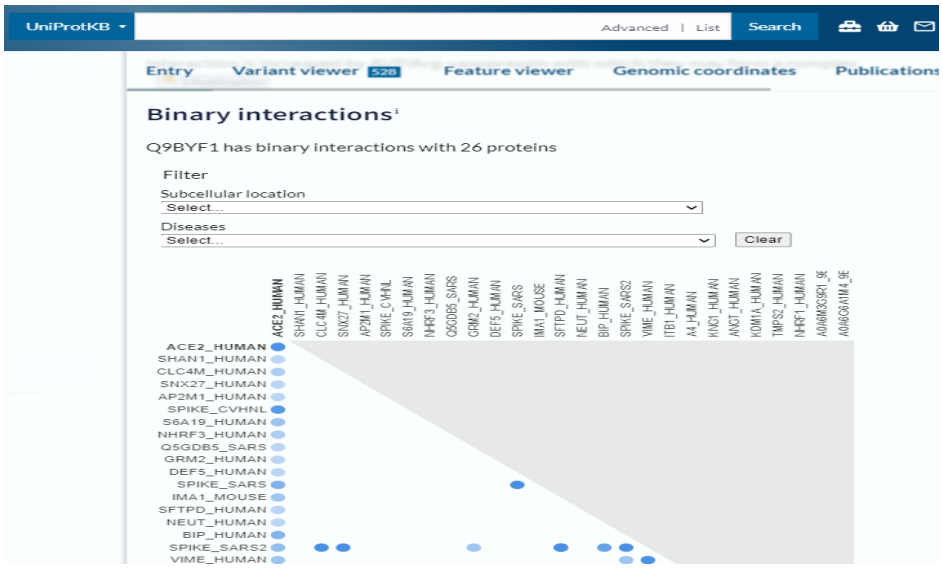


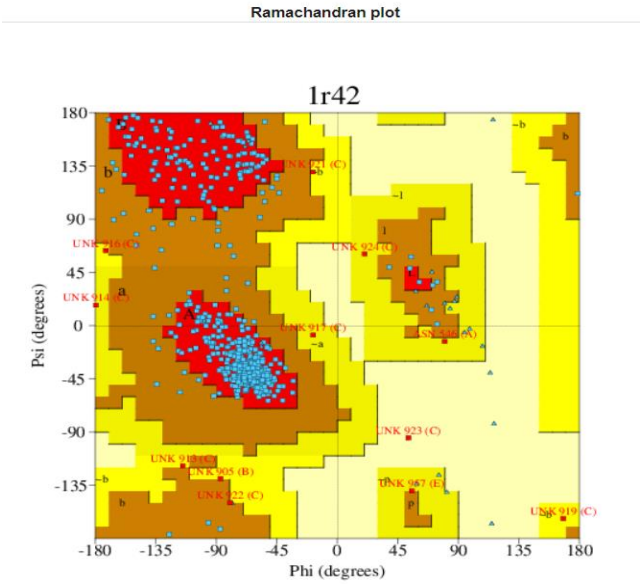
Ribbon Structure of SARS CoV-2 Spike Glycoprotein its analysis section from RCSB PDB



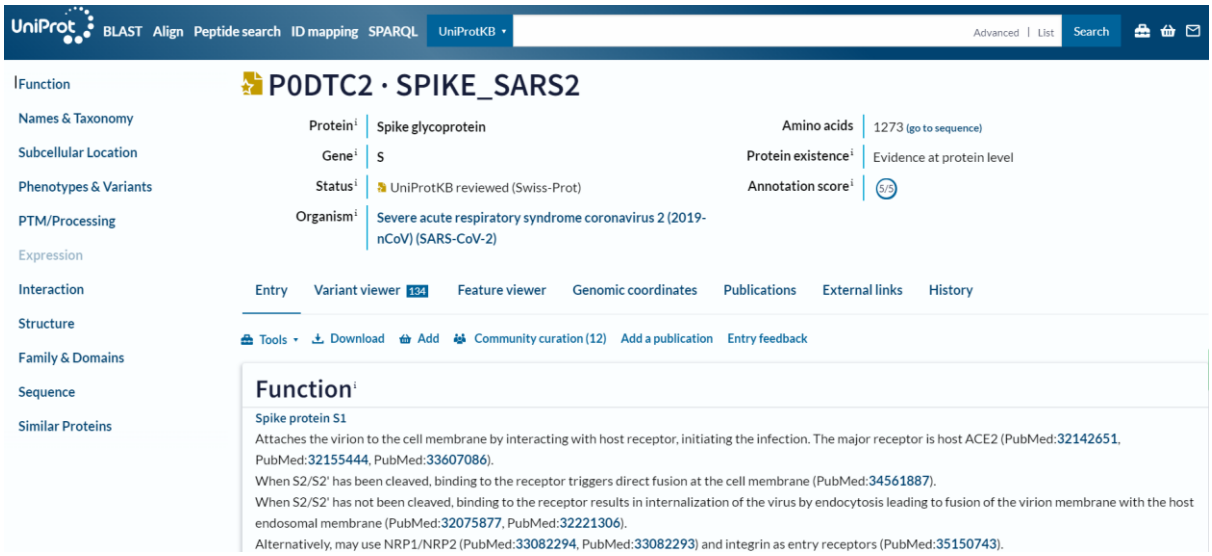
- **UniProtKB Analysis: -**
    - To analyze the interaction of SARS-CoV-2 spike glycoprotein with human ACE2 in UniProtKB
    - Go to the UniProtKB database and enter the UniProt ID **P0DTC2** for the spike protein and **Q9BYF1** for ACE2. For both proteins, look at the functional annotations, domain structures, and interaction data.
    - Cross-reference PDB entries to view structural details. Further scrutiny could also be obtained directly through the use of PDBsum on UniProt and generation of Ramachandran plots as well as review of the binding region.







- The Ramachandran plot for ACE2 (PDB ID: 1R42) shows the distribution of phi (ϕ) and psi (ψ) angles of residues, indicating their secondary structures. Beta sheets are represented by points in the top-left quadrant (ϕ ≈ -135°, ψ ≈ +135°), reflecting the β-sheet conformation. Alpha helices are clustered in the bottom-left quadrant (ϕ ≈ -60°, ψ ≈ -45°), corresponding to α-helix conformations. The plot shows a dense population in both regions, indicating the presence of these secondary structures in ACE2. The colored regions denote allowed and favored angles, suggesting overall structural stability and proper folding.

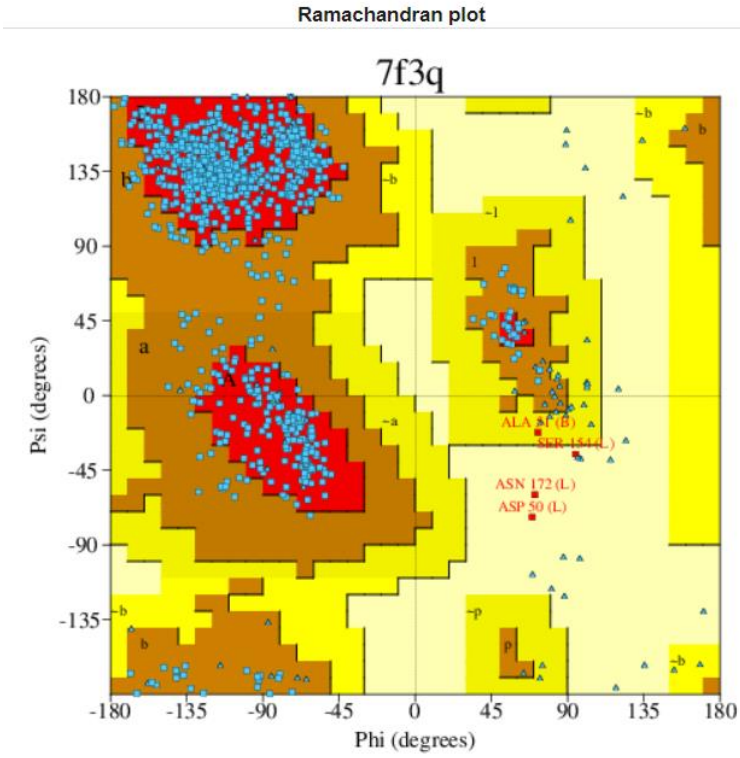


Spike glycoprotein

Homotrimer; each monomer consists of a S1 and a S2 subunit (PubMed:32075877, PubMed:32155444, PubMed:32245784).
The resulting peplomers protrude from the virus surface as spikes (PubMed:32979942).
Interacts with ORF3a protein and ORF7a protein (By similarity) (PubMed:32075877, PubMed:32155444, PubMed:32245784, PubMed:32979942).
There are an average of 26 +/-15 spike trimers at the surface of virion particles (PubMed:32979942).
Binds to host MBL2 (PubMed:35102342).
This binding occurs via glycans and inhibits viral infectivity. Inhibition is effective against alpha, beta, gamma, and delta variants (PubMed:35102342).
UniRule Annotation | 5 Publications

Spike protein S1

Binds to host ACE2 (PubMed:32075877, PubMed:32132184, PubMed:32155444, PubMed:32221306, PubMed:32225175, PubMed:32225176, PubMed:33607086).
RBD also interacts with the N-linked glycan on 'Asn-90' of ACE2 (PubMed:33607086).
Cleavage of S generates a polybasic C-terminal sequence on S1 that binds to host Neuropilin-1 (NRP1) and Neuropilin-2 (NRP2) receptors (PubMed:33082293, PubMed:33082294).
Interacts with host integrin alpha-5/beta-1 (ITGA5:ITGB1) and with ACE2 in complex with integrin alpha-5/beta-1 (ITGA5:ITGB1) (PubMed:33102950).
May interact via cytoplasmic c-terminus with M protein (PubMed:33229438).
May interact (via N-terminus) with host bilirubin and biliverdin, thereby preventing antibody binding to the SARS-CoV-2 spike NTD via an allosteric mechanism (PubMed:33888467).
May interact with host LRRC15, thereby allowing attachment to host cells (PubMed:36735681). 13 Publications



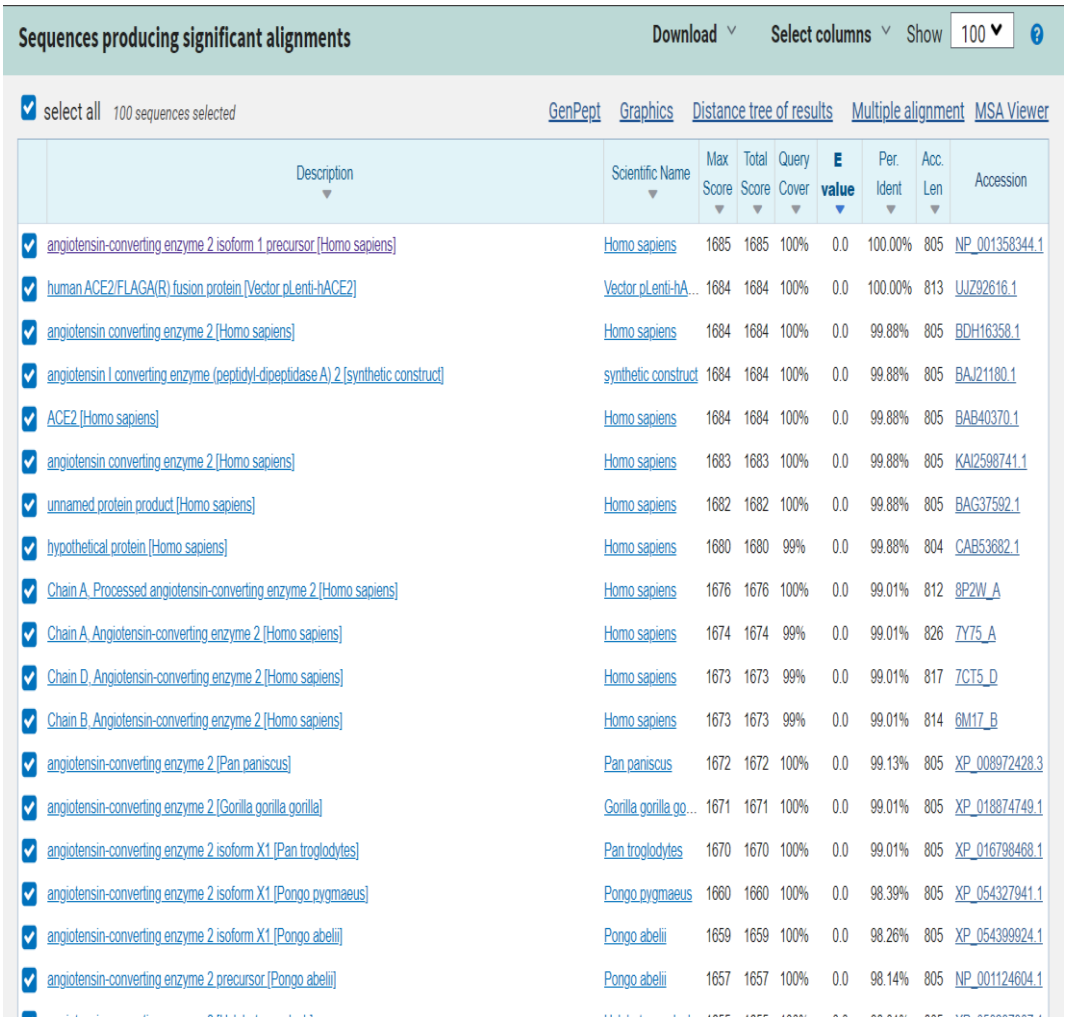
- The majority of points in the top-left quadrant for the SARS-CoV-2 spike glycoprotein (PDB ID: 7F3Q) indicate the presence of beta sheets, which are pointed out by a huge number of points at $\phi \approx -135°$ and $\psi \approx +135°$. Points in the lower part of the left quadrant represent alfa helices at $\phi \approx -60°$, $\psi \approx -45°$. The population of the two regions is well balanced in the plot and implies the spike protein has a balanced composition between β-sheets and α-helices. These favorable regions in the plot indicate a stable protein, which would be crucial for the function of binding to human ACE2 while the virus entry.

(ii). Sequence Alignment and Similarity Searches

- **BLAST**: The Basic Local Alignment Search Tool (BLAST) was utilized for sequence alignment and similarity searches. This tool compared the Spike protein and ACE2 sequences against various databases to identify conserved regions and potential binding sites.

- Go to BLAST on NCBI,
- Upload sequences, and run alignment against databases.
- In comparison with homologous proteins, analyze conserved regions and binding sites of SARS-CoV-2 Spike and ACE2.
- In BLAST, the value of **E-value** or Expect value gives the statistical significance of the match between your query sequence and database sequences. E-value indicates the number of hits expected by chance. Lower values mean greater significance. If an E-value is **0**, it suggests that there is a very strong match, which means that the similarity in the sequences is not due to chance alone.

BLAST Search Results of Human ACE2 Protein Receptor

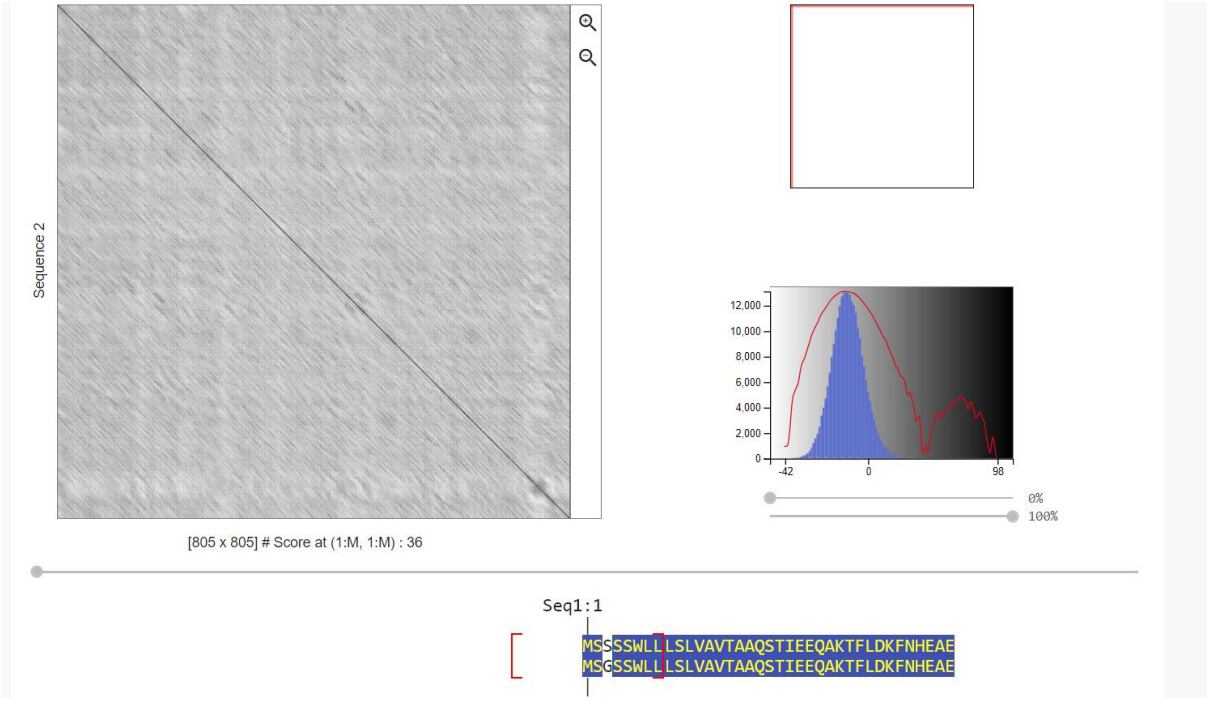


Sequences producing significant alignments Download Select columns Show 100

select all *100 sequences selected* GenPept Graphics Distance tree of results Multiple alignment MSA Viewer

Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
angiotensin-converting enzyme 2 isoform 1 precursor [Homo sapiens]	Homo sapiens	1685	1685	100%	0.0	100.00%	805	NP_001358344.1
human ACE2/FLAGA(R) fusion protein [Vector pLenti-hACE2]	Vector pLenti-hA...	1684	1684	100%	0.0	100.00%	813	UJZ92616.1
angiotensin converting enzyme 2 [Homo sapiens]	Homo sapiens	1684	1684	100%	0.0	99.88%	805	BDH16358.1
angiotensin I converting enzyme (peptidyl-dipeptidase A) 2 [synthetic construct]	synthetic construct	1684	1684	100%	0.0	99.88%	805	BAJ21180.1
ACE2 [Homo sapiens]	Homo sapiens	1684	1684	100%	0.0	99.88%	805	BAB40370.1
angiotensin converting enzyme 2 [Homo sapiens]	Homo sapiens	1683	1683	100%	0.0	99.88%	805	KAI2598741.1
unnamed protein product [Homo sapiens]	Homo sapiens	1682	1682	100%	0.0	99.88%	805	BAG37592.1
hypothetical protein [Homo sapiens]	Homo sapiens	1680	1680	99%	0.0	99.88%	804	CAB53682.1
Chain A, Processed angiotensin-converting enzyme 2 [Homo sapiens]	Homo sapiens	1676	1676	100%	0.0	99.01%	812	8P2W_A
Chain A, Angiotensin-converting enzyme 2 [Homo sapiens]	Homo sapiens	1674	1674	99%	0.0	99.01%	826	7Y75_A
Chain D, Angiotensin-converting enzyme 2 [Homo sapiens]	Homo sapiens	1673	1673	99%	0.0	99.01%	817	7CT5_D
Chain B, Angiotensin-converting enzyme 2 [Homo sapiens]	Homo sapiens	1673	1673	99%	0.0	99.01%	814	6M17_B
angiotensin-converting enzyme 2 [Pan paniscus]	Pan paniscus	1672	1672	100%	0.0	99.13%	805	XP_008972428.3
angiotensin-converting enzyme 2 [Gorilla gorilla gorilla]	Gorilla gorilla go...	1671	1671	100%	0.0	99.01%	805	XP_018874749.1
angiotensin-converting enzyme 2 isoform X1 [Pan troglodytes]	Pan troglodytes	1670	1670	100%	0.0	99.01%	805	XP_016798468.1
angiotensin-converting enzyme 2 isoform X1 [Pongo pygmaeus]	Pongo pygmaeus	1660	1660	100%	0.0	98.39%	805	XP_054327941.1
angiotensin-converting enzyme 2 isoform X1 [Pongo abelii]	Pongo abelii	1659	1659	100%	0.0	98.26%	805	XP_054399924.1
angiotensin-converting enzyme 2 precursor [Pongo abelii]	Pongo abelii	1657	1657	100%	0.0	98.14%	805	NP_001124604.1

DotPlot result of two sequences of ACE2 protein Variants



Alignment view Pairwise Restore defaults

100 sequences selected

Download GenPept Graphics

angiotensin-converting enzyme 2 isoform 1 precursor [Homo sapiens]

Sequence ID: NP_001358344.1 **Length:** 805 **Number of Matches:** 1

See 34 more title(s) See all Identical Proteins(IPG)

Range 1: 1 to 805 GenPept Graphics Next Match Previous Match

Score	Expect	Method	Identities	Positives	Gaps
1685 bits(4363)	0.0	Compositional matrix adjust.	805/805(100%)	805/805(100%)	0/805(0%)

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Sbjct  1    MSSSSWLLLSLVAVTAAQSTIEEQAKTFLDKFNHEAEDLFYQSSLASWNYNTNITEENVQ  60

Query  61   NMNNAGDKWSAFLKEQSTLAQMYPLQEIQNLTVKLQLQALQQNGSSVLSEDKSKRLNTIL  120
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Sbjct  61   NMNNAGDKWSAFLKEQSTLAQMYPLQEIQNLTVKLQLQALQQNGSSVLSEDKSKRLNTIL  120

Query  121  NTMSTIYSTGKVCNPDNPQECLLLEPGLNEIMANSLDYNERLWAWESWRSEVGKQLRPLY  180
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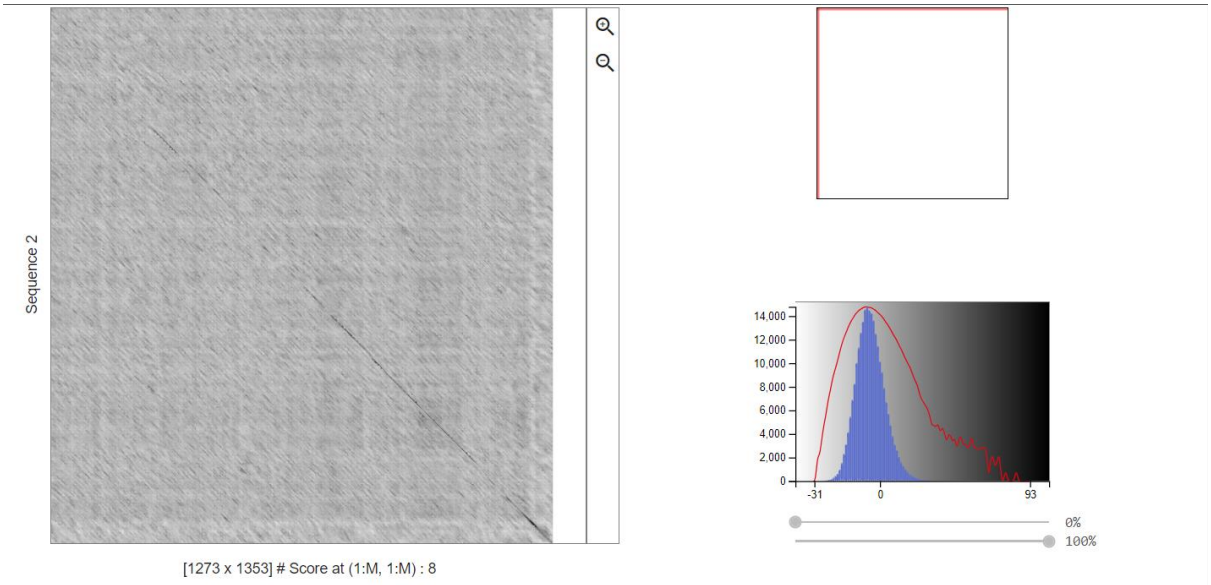
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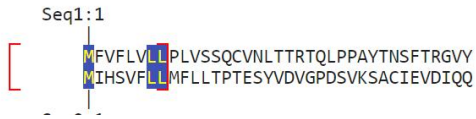
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Sbjct  241  HAYVRAKLMNAYPSYISPIGCLPAHLLGDMWGRFWTNLYSLTVPFGQKPNIDVTDAMVDQ  300

Query  301  AWDAQRIFKEAEKFFVSVGLPNMTQGFWENSMLTDPGNVQKAVCHPTAWDLGKGDFRILM  360
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Sbjct  301  AWDAQRIFKEAEKFFVSVGLPNMTQGFWENSMLTDPGNVQKAVCHPTAWDLGKGDFRILM  360
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BLAST Search Results of Spike Glycoprotein

DotPlot result of two sequences of Spike protein Variants

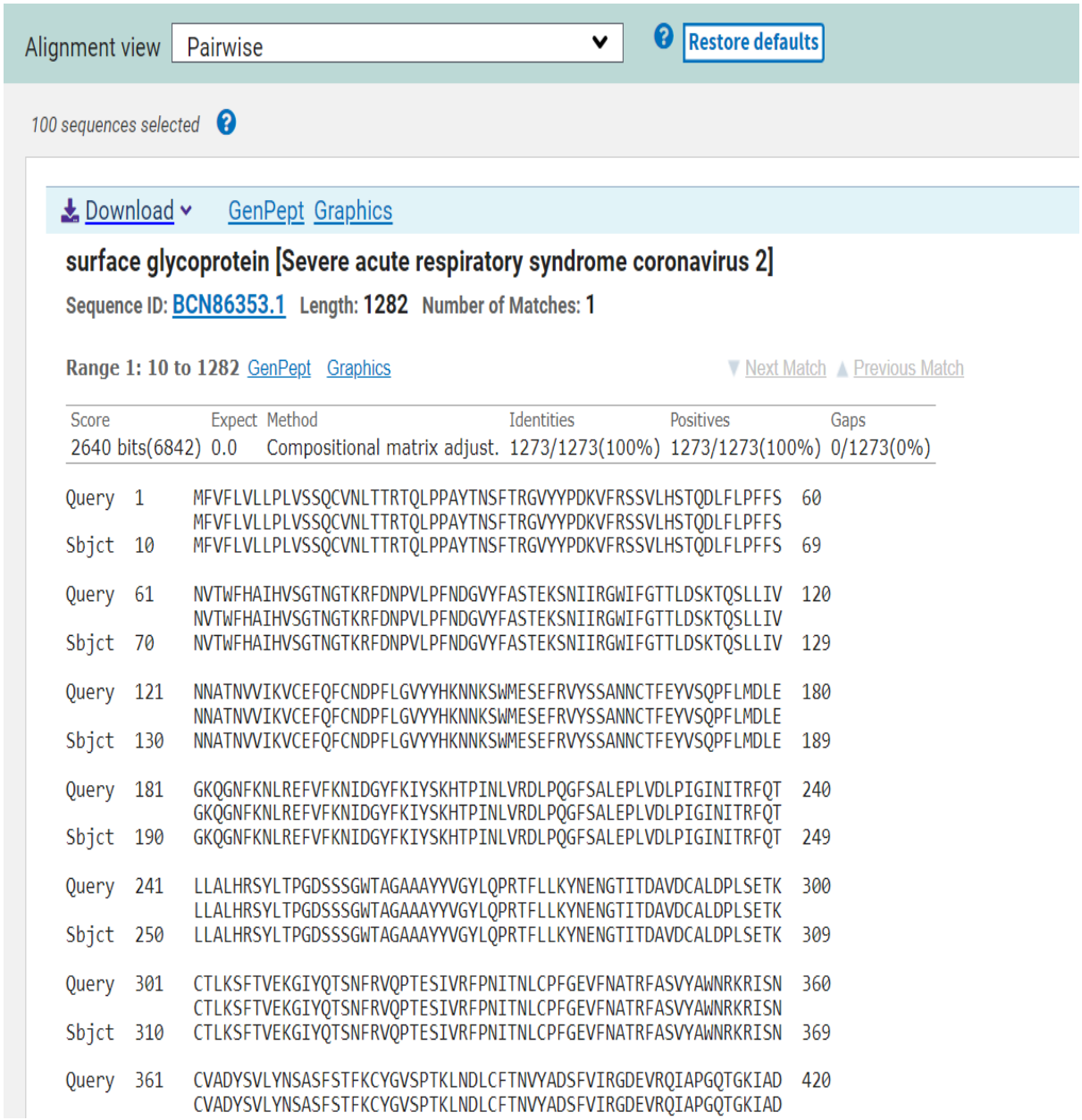




Sequences producing significant alignments Download Select columns Show 100

select all *100 sequences selected* GenPept Graphics Distance tree of results Multiple alignment MSA Viewer

Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
surface glycoprotein [Severe acute respiratory syndrome coronavirus 2]	Severe acute respiratory syndrome coronavir…	2640	2640	100%	0.0	100.00%	1282	BCN86353.1
Chain A, Spike glycoprotein [Severe acute respiratory syndrome coronavirus 2]	Severe acute respiratory syndrome coronavir…	2638	2638	100%	0.0	100.00%	1310	6XR8_A
surface glycoprotein [Severe acute respiratory syndrome coronavirus 2]	Severe acute respiratory syndrome coronavir…	2637	2637	100%	0.0	99.92%	1273	QJF75467.1
Chain C, Spike glycoprotein [Severe acute respiratory syndrome coronavirus 2]	Severe acute respiratory syndrome coronavir…	2637	2637	100%	0.0	100.00%	1281	8DT3_C
surface glycoprotein [Severe acute respiratory syndrome coronavirus 2]	Severe acute respiratory syndrome coronavir…	2637	2637	100%	0.0	99.92%	1273	QJX45031.1
surface glycoprotein [Severe acute respiratory syndrome coronavirus 2]	Severe acute respiratory syndrome coronavir…	2637	2637	100%	0.0	99.92%	1273	QJR85953.1
surface glycoprotein [Severe acute respiratory syndrome coronavirus 2]	Severe acute respiratory syndrome coronavir…	2637	2637	100%	0.0	100.00%	1273	YP_009724390.1
surface glycoprotein [Severe acute respiratory syndrome coronavirus 2]	Severe acute respiratory syndrome coronavir…	2637	2637	100%	0.0	99.92%	1273	QRN64146.1
surface glycoprotein [Severe acute respiratory syndrome coronavirus 2]	Severe acute respiratory syndrome coronavir…	2637	2637	100%	0.0	99.92%	1273	QNN86157.1
surface glycoprotein [Severe acute respiratory syndrome coronavirus 2]	Severe acute respiratory syndrome coronavir…	2637	2637	100%	0.0	99.92%	1273	QJF77846.1
surface glycoprotein [Severe acute respiratory syndrome coronavirus 2]	Severe acute respiratory syndrome coronavir…	2637	2637	100%	0.0	99.92%	1273	QRN78371.1
surface glycoprotein [Severe acute respiratory syndrome coronavirus 2]	Severe acute respiratory syndrome coronavir…	2637	2637	100%	0.0	99.92%	1273	QMS52716.1
surface glycoprotein [Severe acute respiratory syndrome coronavirus 2]	Severe acute respiratory syndrome coronavir…	2636	2636	100%	0.0	99.92%	1273	QIZ16509.1
surface glycoprotein [Severe acute respiratory syndrome coronavirus 2]	Severe acute respiratory syndrome coronavir…	2636	2636	100%	0.0	99.92%	1273	QMI90807.1
surface glycoprotein [Severe acute respiratory syndrome coronavirus 2]	Severe acute respiratory syndrome coronavir…	2636	2636	100%	0.0	99.92%	1273	QIZ97039.1
surface glycoprotein [Severe acute respiratory syndrome coronavirus 2]	Severe acute respiratory syndrome coronavir…	2636	2636	100%	0.0	99.92%	1273	QKS90791.1
surface glycoprotein [Severe acute respiratory syndrome coronavirus 2]	Severe acute respiratory syndrome coronavir…	2636	2636	100%	0.0	99.92%	1273	QQP45825.1
surface glycoprotein [Severe acute respiratory syndrome coronavirus 2]	Severe acute respiratory syndrome coronavir…	2636	2636	100%	0.0	99.92%	1273	QMJ01317.1



- **T-Coffee**: Multiple sequence alignments were performed using T-Coffee to align the SARS-CoV-2 Spike protein with other viral Spike proteins (acting in different organisms) and human ACE2 homologs (means ACE2 receptor in different organisms).

- Use the T-Coffee web server and upload Spike protein and ACE2 homologs.
- Use this tool to conduct multiple sequence alignments identifying conserved domains and mutations relevant to a protein-protein interaction.

T-Coffee alignment result

MSA

The multiple sequence alignment result as produced by T-coffee.

```
T-COFFEE, Version_11.00 (Version_11.00)
Cedric Notredame
SCORE=858
*
 BAD AVG GOOD
*
sp|P0DTC2.1|SPI     :  80
YP 009047204.1      :  76
YP 009724390.1      :  80
YP 009555241.1      :  80
YP 003767.1         :  73
NP 073551.1         :  77
YP 173238.1         :  80
NP_150077.1         :  80
cons                :  85

sp|P0DTC2.1|SPI     MFVF--LVL-LP---LVSS------
QC----VNLTTR-TQ--LPP----AYTN----------
YP 009047204.1      MIHSVFLLM-
FLLTPTESYVDVGPDSVKSACIEVDIQQTF--FDK----
TWPRP---I-----
YP_009724390.1      MFVF--LVL-LP---LVSS------
QC----VNLTTR-TQ--LPP----AYTN----------
YP 009555241.1      MFLI--LLISLP---TAFAVI-
GDLKCT--SDNINDKDTG--PPPI----STDT---V-----
YP 003767.1         MKLF--LIL-LV-LPLASCF----
FTCN-SNANLSMLQLG--VPD----NSST---IVTG-LL
NP 073551.1         MFVL--LVA-YA---LLHI-----
AGCQ--TT--NGLNTSYSVCNGCVGYSENVFAVESGGYI
YP 173238.1         MLLI--IFI-LP---TTLAVI-
GDFNCT--NFAINDLNTT--VPRI----SEYV---V-----
NP_150077.1         MFLI--LLISLP---TAFAVI-
GDLKCT--TVSINDVDTG--VPSI----STDT---V-----

cons                *  :.

sp|P0DTC2.1|SPI     ----------------------
SFTRGVYYPD-KVFRSSVLHSTQDLFLPFFSN---VTWFHAI
YP 009047204.1      ---------------------
DVSKADGIIYPQGRTYSNITI-TYQGLF-PYQGDHGDMYVYSAG
YP 009724390.1      ----------------------
SFTRGVYYPD-KVFRSSVLHSTQDLFLPFFSN---VTWFHAI
YP 009555241.1      ---------------------
DVTNGLGTYYVLDRVYLNTTL-FLNGYY-PTSG-----STYRNM
YP 003767.1         PT-HWFCANQS---------
TSVYSANGFFYIDVGNHRSAFA-LHTGYY-DANQ-----YYI--Y
NP 073551.1         PSDFAF---NNWFLLTNTSSVV--
DGVVRSFQPLLLNCLW-SVSGLR-FTTG-----FVY--F
YP 173238.1         ---------------------
DVSYGLGTYYILDRVYLNTTI-LFTGYF-PKSG-----ANFRDL
NP 150077.1         ---------------------
DVTNGLGTYYVLDRVYLNTTL-LLNGYY-PTSG-----STYRNM

cons                             *     .     .
```

MSA of Viral Spike Glycoprotein Variants

T-Coffee alignment result

MSA

The multiple sequence alignment result as produced by T-coffee.

```
T-COFFEE, Version_11.00 (Version_11.00)
Cedric Notredame
SCORE=1000
*
 BAD AVG GOOD
*
NP_001358344.1    : 100
XP_018874749.1    : 100
XP_016798468.1    : 100
ACI04556.1        : 100
XP_001490241.1    : 100
XP_007090142.2    : 100
NP_001123985.1    : 100
NP_001012006.1    : 100
cons              : 100

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XP_018874749.1    MSGSSWLLLSLVAVTAAQSTIEEQAKTFLDKFNHEAEDLFYQSSLASWNYNTNITEENVQNMNNAG
XP_016798468.1    MSGSSWLLLSLVAVTAAQSTIEEQAKTFLDKFNHEAEDLFYQSSLASWNYNTNITEENVQNMNNAG
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XP_007090142.2    MSGSFWLLLSFAALTAAQSTTEELAKTFLEKFNHEAEELSYQSSLASWNYNTNITDENVQKMNEAG
NP_001123985.1    MSSSSWLLLSLVAVTTAQSLTEENAKTFLNNFNQEAEDLSYQSSLASWNYNTNITEENAQKMSEAA
NP_001012006.1    MSSSCWLLLSLVAVATAQSLIEEKAESFLNKFNQEAEDLSYQSSLASWNYNTNITEENAQKMNEAA

cons              **.* *****:.*:::***  *: *::**:;** ***:* :*******.******:**.*:*.:*.

NP_001358344.1    DKWSAFLKEQSTLAQMYPLQEIQNLTVKLQLQALQQNGSSVLSEDKSKRLNTILNTMSTIYSTGKV
XP_018874749.1    DKWSAFLKEQSTLAQMYPLQEIQNLTIKLQLQALQQNGSSVLSEDKSKRLNTILNTMSTIYSTGKV
XP_016798468.1    DKWSAFLKEQSTLAQMYPLQEIQNLTVKLQLQALQQNGSSVLSEDKSKRLNTILNTMSAIYSTGKV
ACI04556.1        EKWSAFLKEQSTLAQMYPLQEIQNLTVKLQLQALQQNGSSVLSEDKSKRLNTILNTMSTIYSTGKV
XP_001490241.1    ARWSAFYEEQCKLAKTYPLEEIQNLTVKRQLQALQQSGSSVLSADKSKRLNEILNTMSTIYSTGKV
XP_007090142.2    AKWSAFYEEQSKLAETYPLAEIHNTTVKRQLQALQQSGSSVLSADKSQRLNTILNAMSTIYSTGKA
NP_001123985.1    AKWSAFYEEQSKTAQSFSLQEIQTPIIKRQLQALQQSGSSALSADKNKQLNTILNTMSTIYSTGKV
NP_001012006.1    AKWSAFYEEQSKIAQNFSLQEIQNATIKRQLKALQQSGSSALSPDKNKQLNTILNTMSTIYSTGKV

cons              :**** :**.. *: :.* **:.  :* **:****.***.** **.::** ***:**:******.

NP_001358344.1    CNPDNPQECLLLEPGLNEIMANSLDYNERLWAWESWRSEVGKQLRPLYEEYVVLKNEMARANHYED
XP_018874749.1    CNPNNPQECLLLEPGLNEIMANSLDYSERLWAWESWRSEVGKQLRPLYEEYVVLKNEMARANHYED
XP_016798468.1    CNPNNPQECLLLEPGLNEIMANSLDYNERLWAWESWRSEVGKQLRPLYEEYVVLKNEMARANHYED
ACI04556.1        CNPNNPQECLLLDPGLNEIMEKSLDYNERLWAWEGWRSEVGKQLRPLYEEYVVLKNEMARANHYKD
XP_001490241.1    CNPSNPQECLLLEPGLDAIMENSKDYNQRLWAWEGWRSEVGKQLRPLYEEYVVLKNEMARANNYED
XP_007090142.2    CNPNNPQECLLLEPGLDDIMENSKDYNERLWAWEGWRAEVGKQLRPLYEEYVALKNEMARANNYED
NP_001123985.1    CNPKNPQECLLLEPGLDEIMATSTDYNSRLWAWEGWRAEVGKQLRPLYEEYVVLKNEMARANNYND
```

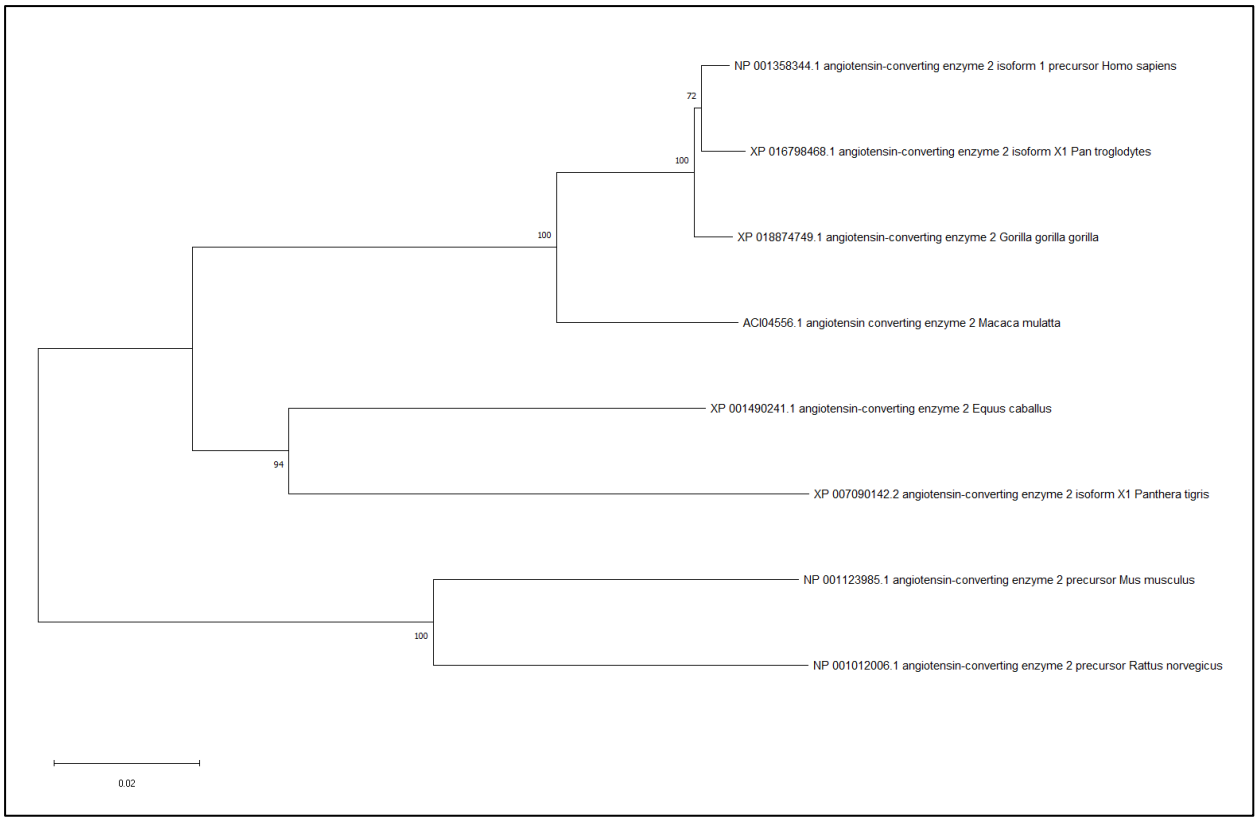
MSA of Human ACE2 Protein Variants

- ***Phylogentic Analysis:***

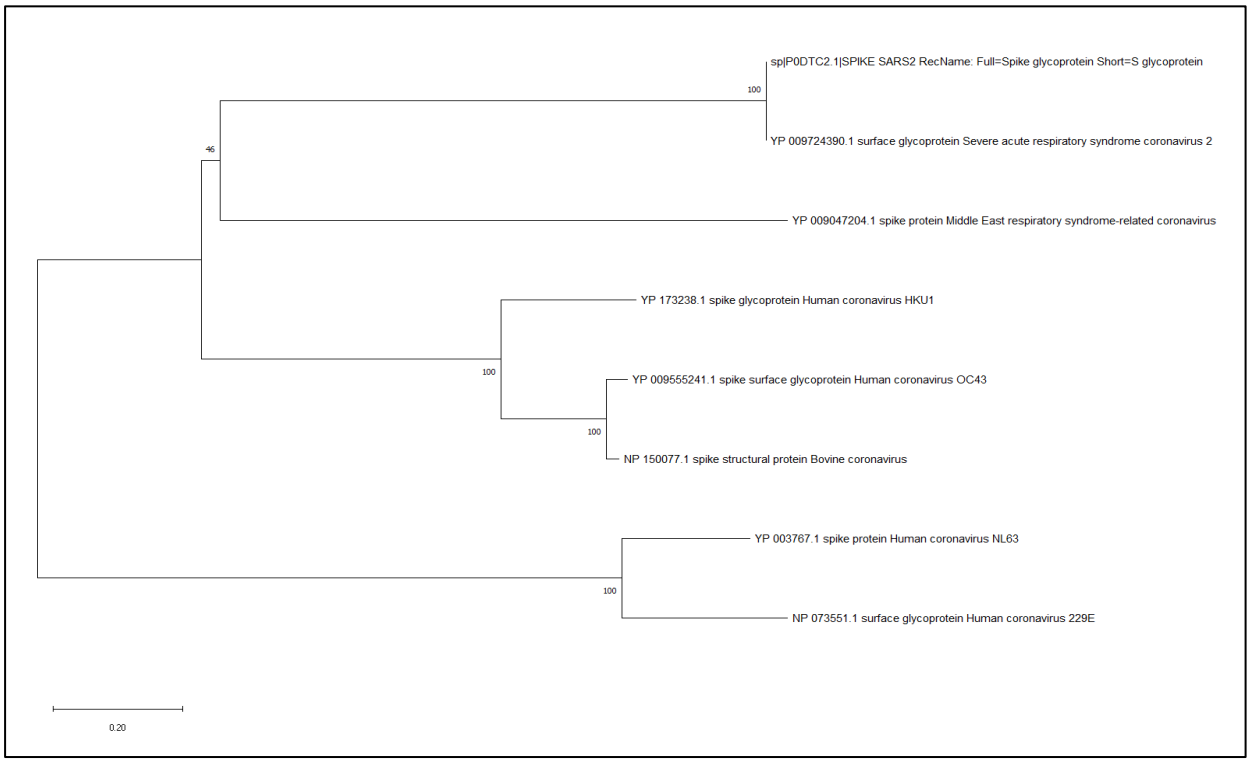
The phylogenetic analysis aimed to understand the evolutionary relationships between the SARS-CoV-2 Spike glycoprotein and its homologs in other coronaviruses, as well as to explore the evolutionary conservation of the ACE2 binding interface.The phylogenetic analysis revealed the close evolutionary relationship between SARS-CoV-2 and other Betacoronaviruses, particularly SARS-CoV.

- Download MEGA XI and input aligned sequences of Spike proteins and ACE2 homologs into it.
- Perform phylogenetic tree construction using *Maximum Likelihood method*, aligned with *MUSCLE*, for protein alignment.
- The phylogenetic tree was constructed with *100 bootstrap replicates* to investigate the evolutionary relationships between the strains, particularly the conservational pattern of Spike-ACE2 binding sites using Maximum Likelihood Method Phylogenetic Tree construction.

Phylogenetic Tree of ACE2 Protein Variants

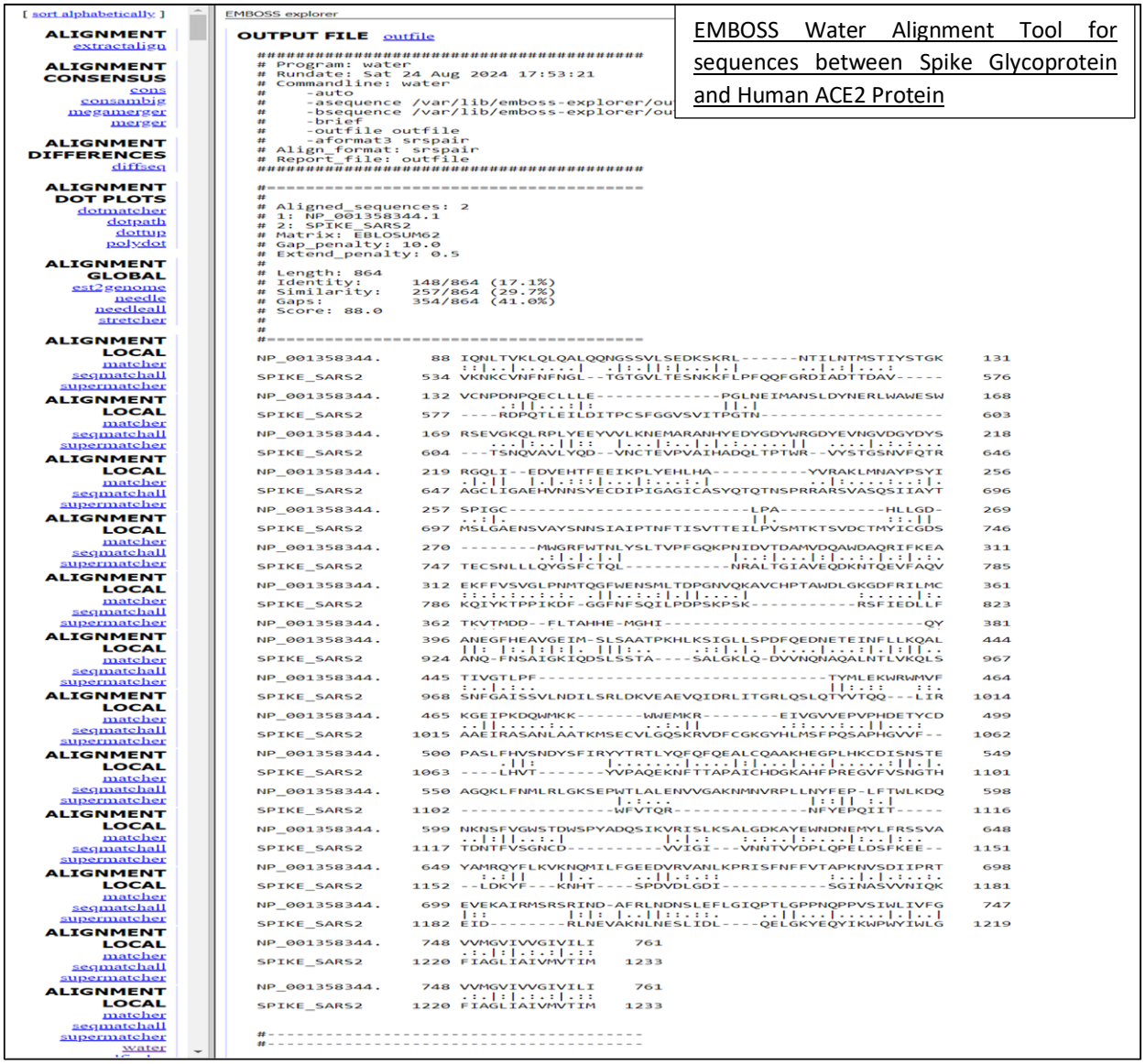


Phylogenetic Tree of Spike Protein Variants



- **EMBOSS Explorer:** For the sequence alignments, I used EMBOSS tools to find out how the spike protein is interacting with ACE2. The important functions in this regard were local pairwise alignment with EMBOSS *Water*, the visualization of conserved regions with *Dotmatcher*, physiochemical properties with *Pepstats*, and antigenic prediction through *Antigenic*.

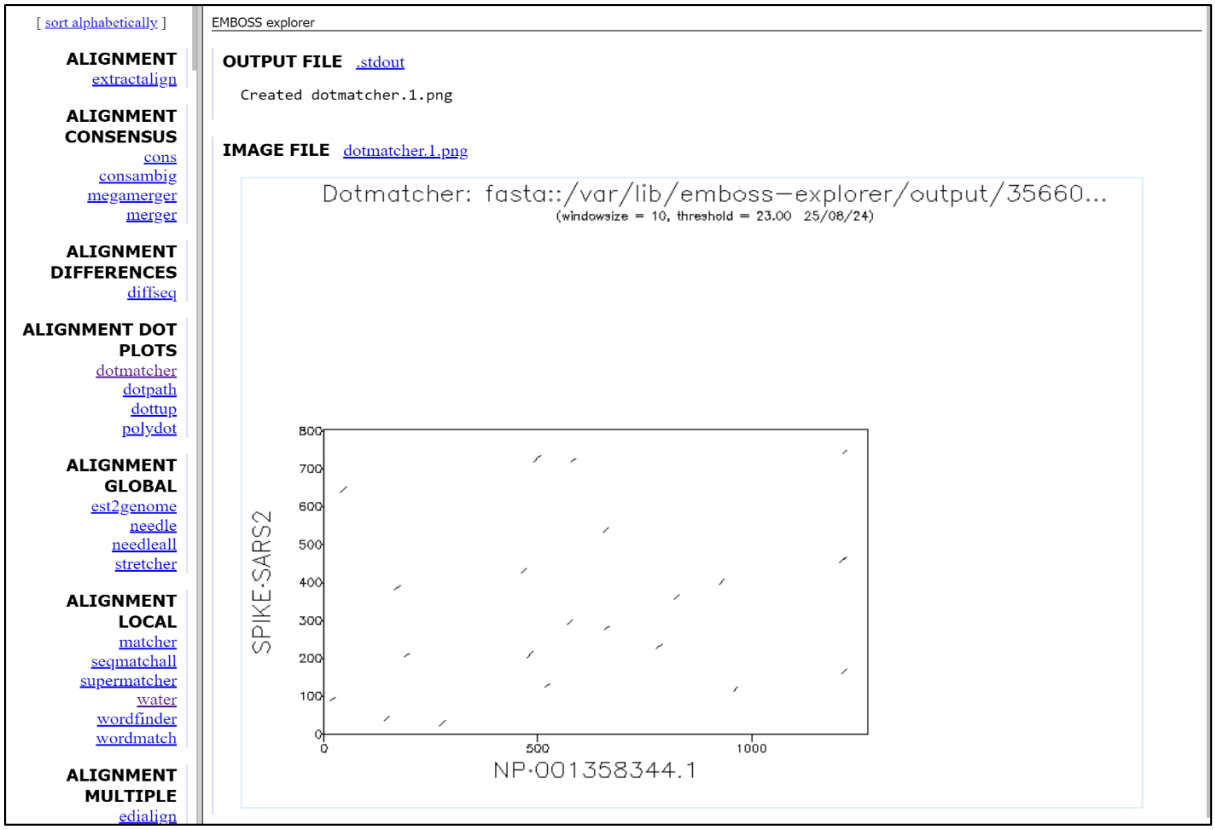
  - Use EMBOSS *Water* for local pairwise alignment
  - *Dotmatcher* to visualize the conserved region,
  - *Pepstats* to perform a physicochemical analysis, and
  - *Antigenic* to predict antigenic regions involved in Spike-ACE2 binding.



EMBOSS Water Alignment Tool for sequences between Spike Glycoprotein and Human ACE2 Protein

This is a local sequence alignment of **SPIKE_SARS2** SARS-CoV-2 spike protein with **NP_001358344** human ACE2 protein from the EMBOSS *Water* tool. With an alignment score of 880.0, it shows a somewhat fair level of similarity. The sequences come to align over 1233 and 805 residues for SPIKE_SARS2 and NP_001358344, respectively, for which gaps were inserted to maximize alignment. The alignment reveals regions of high similarity but many mismatches and gaps, which suggest major structural and functional differences between the two proteins. It has brought out key regions where the viral spike protein may interact with the ACE2, crucial for viral entry into host cells.

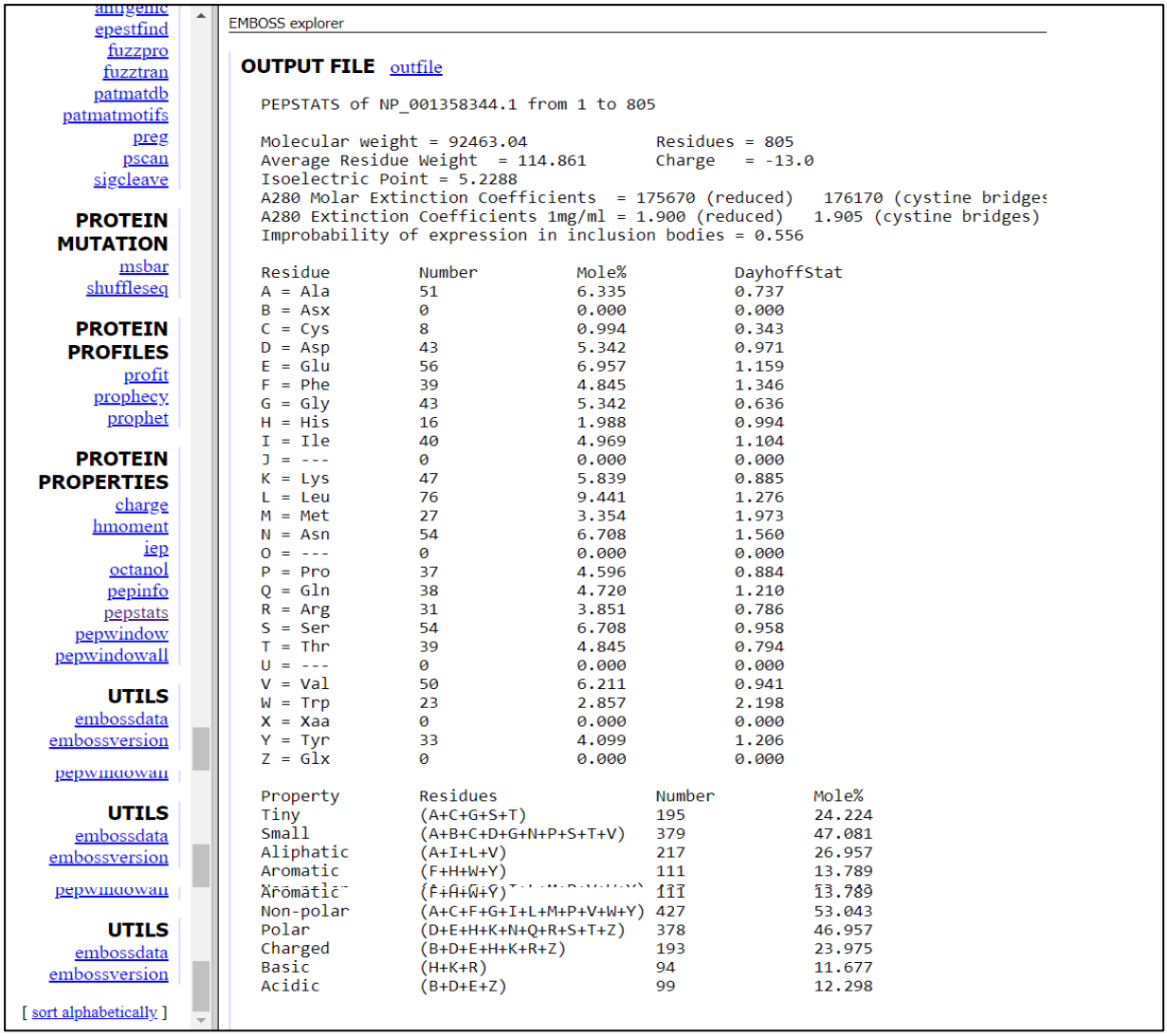
EMBOSS Dotmatcher Tool for sequences of Spike Glycoprotein and Human ACE2 Protein



This image shows the output from the *Dotmatcher* tool in EMBOSS, which is used to visualize sequence similarities between two proteins, in this case, **SPIKE_SARS2** (SARS-CoV-2 spike protein) and **NP_001358344.1** (human ACE2). The dot plot reveals regions of alignment, where diagonal lines indicate local matches between the two sequences. The sparsity of these diagonal segments suggests limited regions of high similarity between the spike protein and ACE2. This aligns with the expectation that while these proteins interact during viral entry, they are structurally and functionally distinct overall. The plot helps identify specific alignment regions for further investigation.

EMBOSS Pepstats Tool for the sequence of Human ACE2 Protein

This image presents the output from *PEPSTATS* in EMBOSS, showing a physicochemical analysis of the **NP_001358344.1** sequence (human ACE2 protein). Key metrics include a molecular weight of 92,463.04 Da, an isoelectric point of 5.23, and a net charge of -13. The analysis also lists amino acid composition, showing a high content of **Leu** (47 residues, 5.84%) and **Ser** (40 residues, 4.97%). Extinction coefficients are given for reduced and cystine-bridged states, useful for protein quantification. The protein is predominantly polar (46.95%), non-polar (53.04%), and moderately acidic, which may influence its interaction with the SARS-CoV-2 spike protein.



antigenic
epestfind
fuzzpro
fuzztran
patmatdb
patmatmotifs
preg
pscan
sigcleave

PROTEIN MUTATION
msbar
shuffleseq

PROTEIN PROFILES
profit
prophecy
prophet

PROTEIN PROPERTIES
charge
hmoment
iep
octanol
pepinfo
pepstats
pepwindow
pepwindowall

UTILS
embossdata
embossversion
pepwindowall

UTILS
embossdata
embossversion
pepwindowall

UTILS
embossdata
embossversion

[sort alphabetically]

EMBOSS explorer

OUTPUT FILE outfile

PEPSTATS of NP_001358344.1 from 1 to 805

Molecular weight = 92463.04 Residues = 805
Average Residue Weight = 114.861 Charge = -13.0
Isoelectric Point = 5.2288
A280 Molar Extinction Coefficients = 175670 (reduced) 176170 (cystine bridges
A280 Extinction Coefficients 1mg/ml = 1.900 (reduced) 1.905 (cystine bridges)
Improbability of expression in inclusion bodies = 0.556

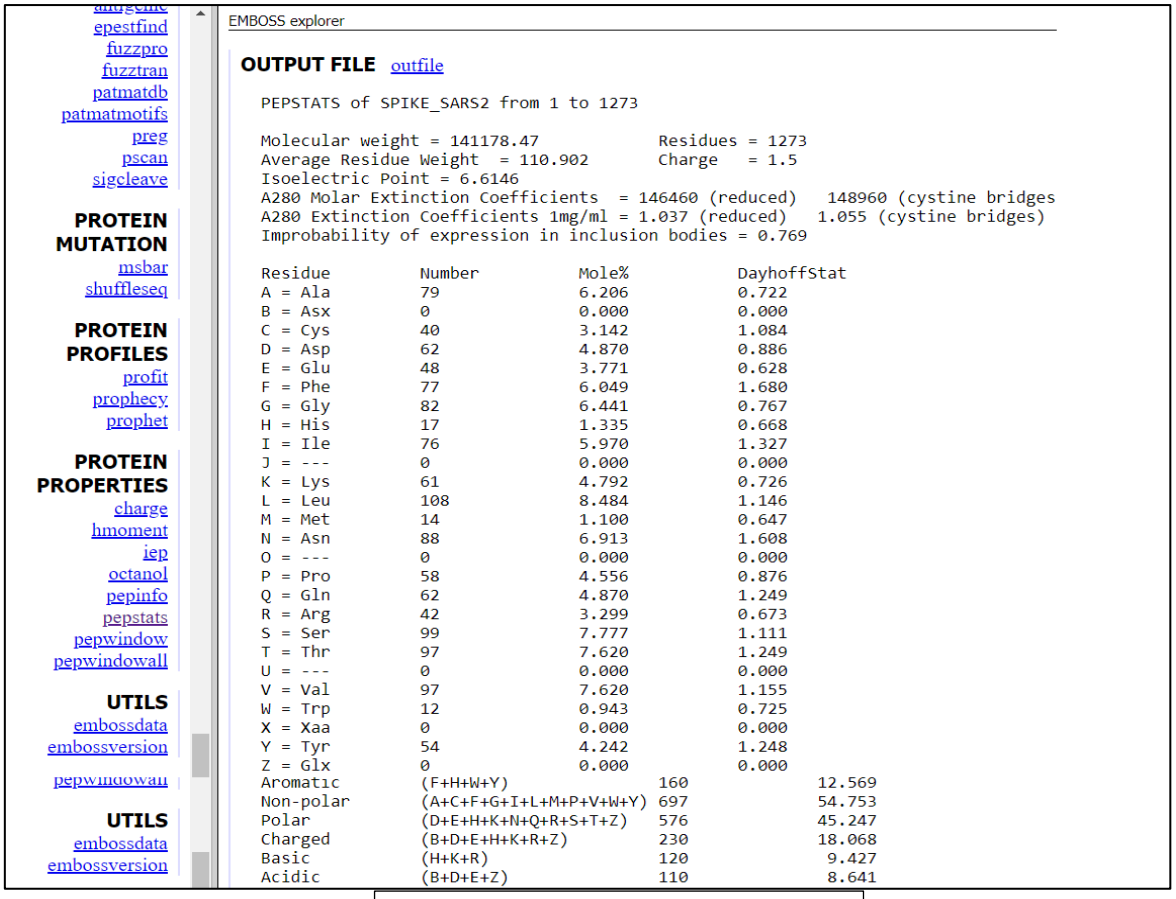
Residue	Number	Mole%	DayhoffStat
A = Ala	51	6.335	0.737
B = Asx	0	0.000	0.000
C = Cys	8	0.994	0.343
D = Asp	43	5.342	0.971
E = Glu	56	6.957	1.159
F = Phe	39	4.845	1.346
G = Gly	43	5.342	0.636
H = His	16	1.988	0.994
I = Ile	40	4.969	1.104
J = ---	0	0.000	0.000
K = Lys	47	5.839	0.885
L = Leu	76	9.441	1.276
M = Met	27	3.354	1.973
N = Asn	54	6.708	1.560
O = ---	0	0.000	0.000
P = Pro	37	4.596	0.884
Q = Gln	38	4.720	1.210
R = Arg	31	3.851	0.786
S = Ser	54	6.708	0.958
T = Thr	39	4.845	0.794
U = ---	0	0.000	0.000
V = Val	50	6.211	0.941
W = Trp	23	2.857	2.198
X = Xaa	0	0.000	0.000
Y = Tyr	33	4.099	1.206
Z = Glx	0	0.000	0.000

Property	Residues	Number	Mole%
Tiny	(A+C+G+S+T)	195	24.224
Small	(A+B+C+D+G+N+P+S+T+V)	379	47.081
Aliphatic	(A+I+L+V)	217	26.957
Aromatic	(F+H+W+Y)	111	13.789
Aromatic	(F+H+W+Y)	111	13.789
Non-polar	(A+C+F+G+I+L+M+P+V+W+Y)	427	53.043
Polar	(D+E+H+K+N+Q+R+S+T+Z)	378	46.957
Charged	(B+D+E+H+K+R+Z)	193	23.975
Basic	(H+K+R)	94	11.677
Acidic	(B+D+E+Z)	99	12.298

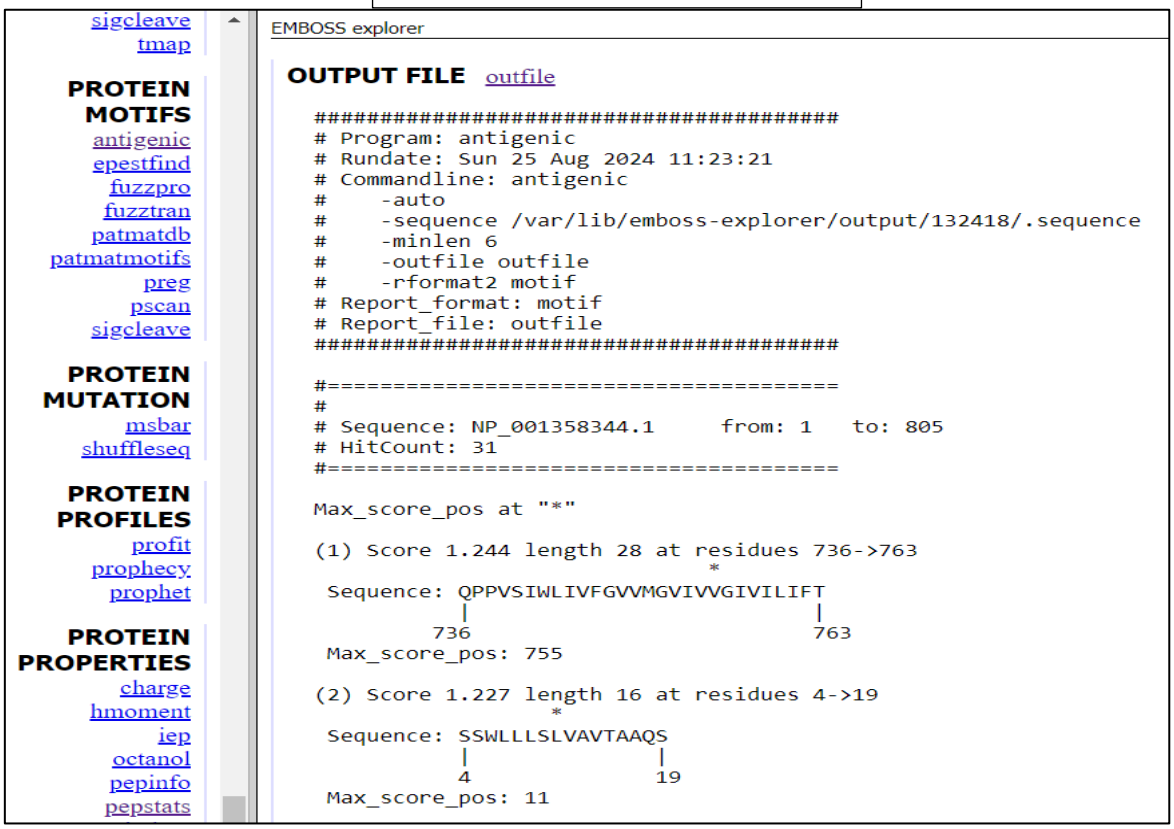
EMBOSS Pepstats Tool for the sequence of SARS CoV-2 Spike Glycoprotein

The EMBOSS Pepstats tool provides essential information about protein sequences, and in this case, it has been applied to the SARS CoV-2 Spike Glycoprotein. Here's a concise summary of the key findings:

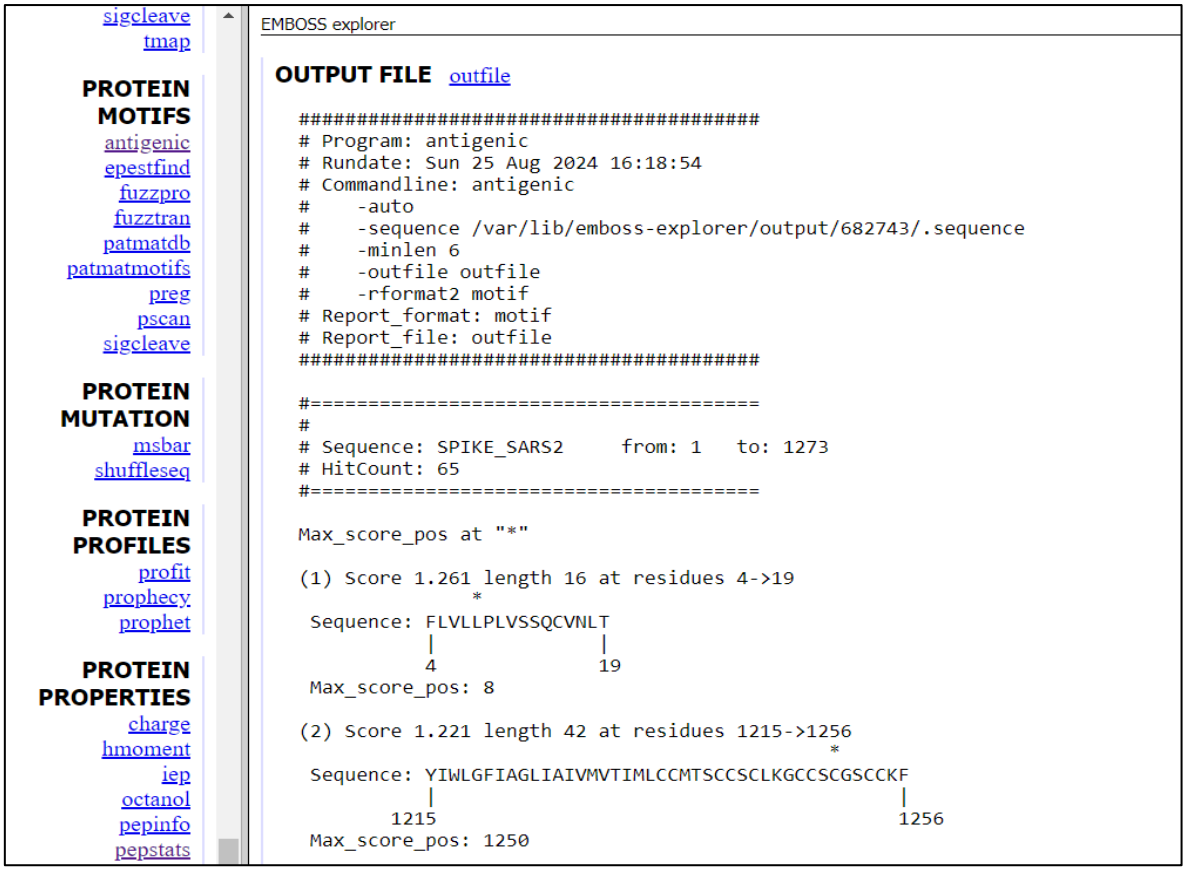
1. **Molecular Weight**: The molecular weight of the Spike Glycoprotein is approximately 141,178.47 Daltons.
2. **Residues**: The protein consists of 1,273 amino acid residues.
3. **Average Residue Weight**: On average, each amino acid residue weighs about 110.902 Daltons.
4. **Isoelectric Point (pI)**: The pI of the protein is 6.6146, which indicates its charge neutrality.



EMBOSS Antigenic Tool for the sequence of Human ACE2 Protein



EMBOSS Antigenic Tool for the sequence of SARS CoV-2 Spike Glycoprotein



The EMBOSS Antigenic Tool analysis of the Human ACE2 protein reveals key antigenic sites that may trigger an immune response. These regions act as potential "hotspots" for antibody binding, providing insights crucial for vaccine development and understanding immune interactions with ACE2. Identifying these sites could help in designing strategies to modulate immune responses, especially in the context of SARS-CoV-2 infection.

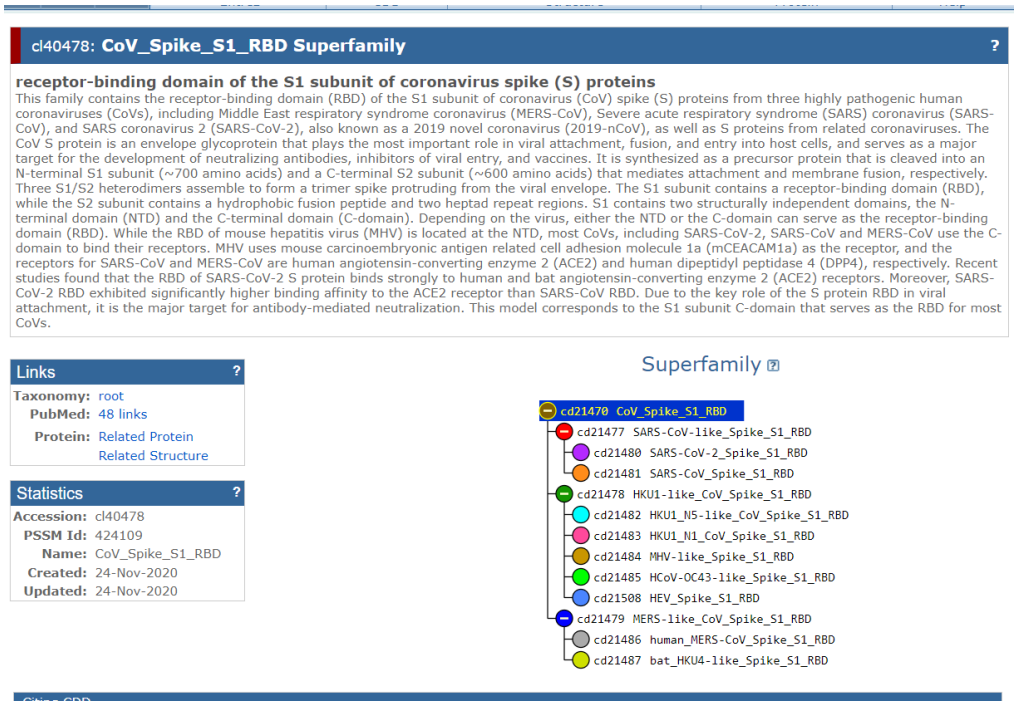
Similarly, the analysis of the SARS-CoV-2 Spike protein (S protein) highlights critical antigenic regions where the immune system may recognize the virus. Understanding these sites aids researchers in crafting targeted vaccines and therapies, effectively pinpointing the virus's vulnerabilities. Both analyses emphasize the importance of these antigenic sites in guiding the development of interventions against COVID-19 and enhancing our understanding of immune mechanisms at play.

(iii). Domain and Motif Analysis

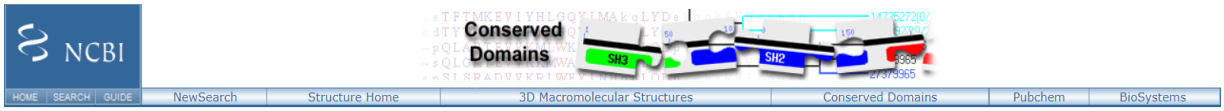
- Conserved Domains Database (CDD) and MOTIF: Domain and motif analysis was conducted using CDD and MOTIF to identify conserved domains and sequence motifs within the Spike glycoprotein and ACE2 that might be involved in the interaction.

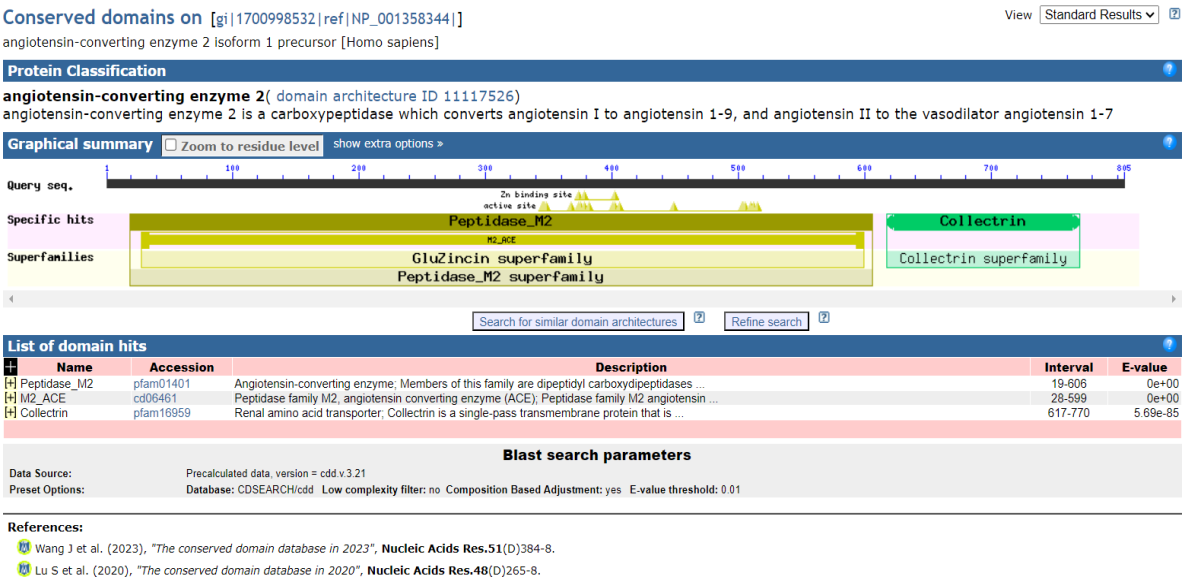
- Access Conserved Domain Database (CDD) and MOTIF.

- Upload the sequences to identify conserved domains and motifs which can participate in binding or are important for structural integrity of both Spike glycoprotein and ACE2.

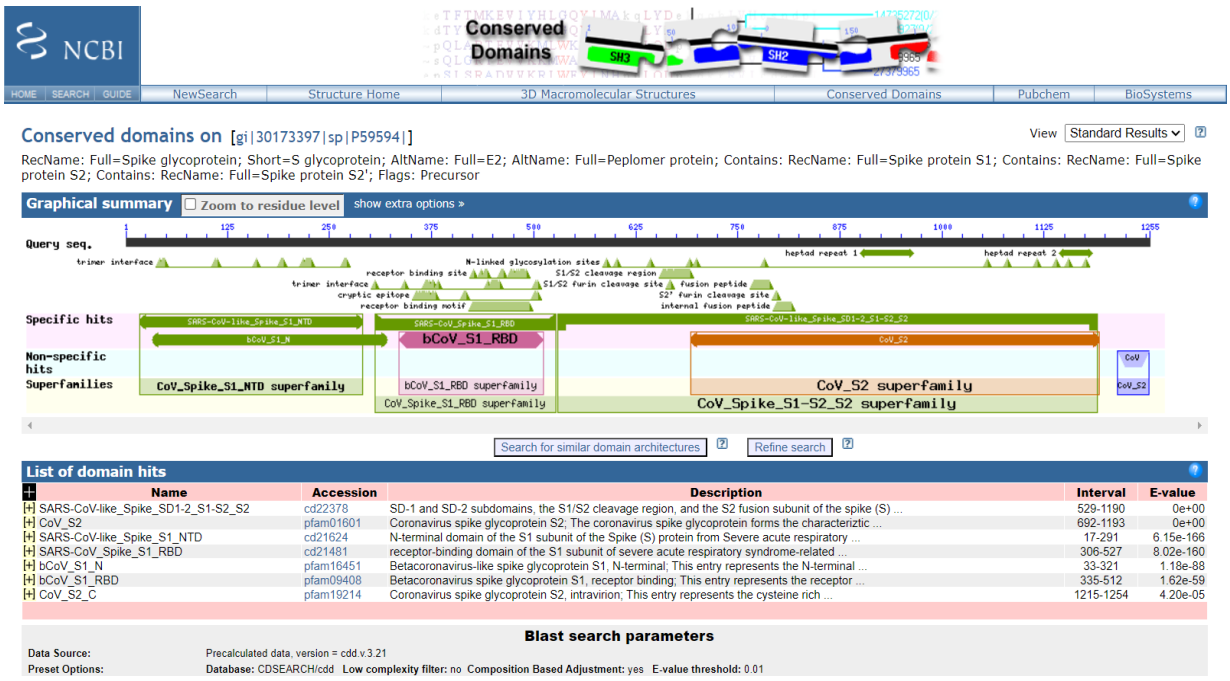


CDD Result of Human ACE2 Protein





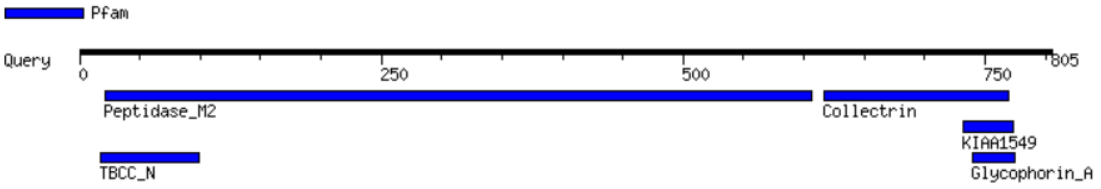
CDD OF SARS CoV-2 Spike Glycoprotein



MOTIF Data Result of Human ACE2 Protein

Result of MotifFinder

Number of found motifs: 5



Pfam (5 motifs)

Pfam	Position(Independent E-value)		Description
Peptidase_M2	21..606(3.8e-250)	Detail	PF01401, Angiotensin-converting enzyme
Collectrin	617..770(6.6e-73)	Detail	PF16959, Renal amino acid transporter
KIAA1549	732..773(0.12)	Detail	PF12877, UPF0606 protein KIAA1549
Glycophorin_A	740..774(0.42)	Detail	PF01102, Glycophorin A
TBCC_N	17..99(0.89)	Detail	PF16752, Tubulin-specific chaperone C N-terminal domain

Motif in the sequence

Pfam ID:
Peptidase_M2
Description:
PF01401, Angiotensin-converting enzyme
Appearance:

Position	21..606
Alignment Query Database	IEEQAKTFLDKFNHEAEDLFYQSSLASWNYNTNITEENVQNMNNAGDKWSAFLKEQSTLAQMYPLQEIQNLTVKLQLQALQQNGSSVLSEDKSKF deaeakefleelekeaekvlnesteaswnyntniteenkqakleaelelaaftkelakkakkfdleefqdedlkrqlkklsklgaaalseeklee
i-Evalue	3.8e-250

Sequence:

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EEYVVLKNEMARANHYEDYGDYWRGDYEVNGVDGYDYSRGQLIEDVEHTFEEIKPLYEHL
HAYVRAKLMNAYPSYISPIGCLPAHLLGDMWGRFWTNLYSLTVPFGQKPNIDVTDAMVDQ
AWDAQRIFKEAEKFFVSVGLPNMTQGFWENSMLTDPGNVQKAVCHPTAWDLGKGDFRILM
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KREIVGVVEPVPHDETYCDPASLFHVSNDYSFIRYYTRTLYQFQFQEALCQAAKHEGPLH
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QMILFGEEDVRVANLKPRISFNFFVTAPKNVSDIIPRTEVEKAIRMSRSRINDAFRLNDN
SLEFLGIQPTLGPPNQPPVSIWLIVFGVVMGVIVVGIVILIFTGIRDRKKKNKARSGENP
YASIDISKGENNPGFQNTDDVQTSF

Motif in the sequence

Pfam ID:
Collectrin
Description:
PF16959, Renal amino acid transporter
Appearance:

Position	617..770
Alignment Query Database	SIKVRISLKSALGDKAYEWNDNEMYLFRSSVAYAMRQYFLKVKNQMILFGEEDVRVANLKPRISFNFFVTAPKNVSDIIPRTEVEKAIRMSRSRINDAFRLNDNSLEFLGIQPTLGPPNQPPVSIWLIVFGVVMGVIVVGIVILIFTGIRDRKK aikvRISlktAlGdkAyewneneeylfkasvayalrkylskeknetvefevenvlvcnetkRvSfwfvvtspknksklipkeeveeairksreriNeaflddktLeflgilptLappveppvtvWlivfgvvlgvvvvgivllivsgirerkr
i-Evalue	6.6e-73

Sequence:

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QMILFGEEDVRVANLKPRISFNFFVTAPKNVSDIIPRTEVEKAIRMSRSRINDAFRLNDN
SLEFLGIQPTLGPPNQPPVSIWLIVFGVVMGVIVVGIVILIFTGIRDRKKKNKARSGENP
YASIDISKGENNPGFQNTDDVQTSF

Motif in the sequence

Pfam ID:
KIAA1549
Description:
PF12877, UPF0606 protein KIAA1549
Appearance:

Position	732..773
Alignment Query Database	GPPNQPPVSIWLIVfGVVMGVIVVGIVILIFTGIRDRKKKNK nppsdtnnnlWiiv.gvvvPilvvvvviiiilyaklCrkdkle
i-Evalue	0.12

Sequence:
MSSSSWLLLSLVAVTAAQSTIEEQAKTFLDKFNHEAEDLFYQSSLASWNYNTNITEENVQ
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NTMSTIYSTGKVCNPDNPQECLLLEPGLNEIMANSLDYNERLWAWESWRSEVGKQLRPLY
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HAYVRAKLMNAYPSYISPIGCLPAHLLGDMWGRFWTNLYSLTVPFGQKPNIDVTDAMVDQ
AWDAQRIFKEAEKFFVSVGLPNMTQGFWENSMLTDPGNVQKAVCHPTAWDLGKGDFRILM
CTKVTMDDFLTAHHEMGHIQYDMAYAAQPFLLRNGANEGFHEAVGEIMSLSAATPKHLKS
IGLLSPDFQEDNETEINFLLKQALTIVGTLPFTYMLEKWRWMVFKGEIPKDQWMKKWWEM
KREIVGVVEPVPHDETYCDPASLFHVSNDYSFIRYYTRTLYQFQFQEALCQAAKHEGPLH
KCDISNSTEAGQKLFNMLRLGKSEPWTLALENVVGAKNMNVRPLLNYFEPLFTWLKDQNK
NSFVGWSTDWSPYADQSIKVRISLKSALGDKAYEWNDNEMYLFRSSVAYAMRQYFLKVKN
QMILFGEEDVRVANLKPRISFNFFVTAPKNVSDIIPRTEVEKAIRMSRSRINDAFRLNDN
SLEFLGIQPTLGPPNQPPVSIWLIVFGVVMGVIVVGIVILIFTGIRDRKKKNKARSGENP
YASIDISKGENNPGFQNTDDVQTSF

Motif in the sequence

Pfam ID:
Glycophorin_A
Description:
PF01102, Glycophorin A
Appearance:

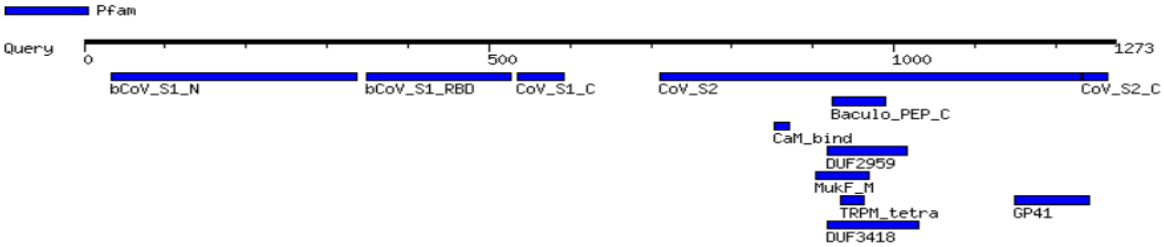
Position	740..774
Alignment Query Database	SIWLIVFGVVMGVIvvGIVILIFTGIRDRKKKNKA vitliifgvmAgvi..gtillisYlirRlikkssv
i-Evalue	0.42

Sequence:
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NTMSTIYSTGKVCNPDNPQECLLLEPGLNEIMANSLDYNERLWAWESWRSEVGKQLRPLY
EEYVVLKNEMARANHYEDYGDYWRGDYEVNGVDGYDYSRGQLIEDVEHTFEEIKPLYEHL
HAYVRAKLMNAYPSYISPIGCLPAHLLGDMWGRFWTNLYSLTVPFGQKPNIDVTDAMVDQ
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KREIVGVVEPVPHDETYCDPASLFHVSNDYSFIRYYTRTLYQFQFQEALCQAAKHEGPLH
KCDISNSTEAGQKLFNMLRLGKSEPWTLALENVVGAKNMNVRPLLNYFEPLFTWLKDQNK
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SLEFLGIQPTLGPPNQPPVSIWLIVFGVVMGVIVVGIVILIFTGIRDRKKKNKARSGENP
YASIDISKGENNPGFQNTDDVQTSF

MOTIF Data Result of SARS CoV-2 Spike Glycoprotein

Result of MotifFinder

Number of found motifs: 12



Pfam (12 motifs)

Pfam	Position(Independent E-value)		Description
CoV_S2	711..1232(2.5e-201)	Detail	PF01601, Coronavirus spike glycoprotein S2
bCoV_S1_N	33..337(6.8e-70)	Detail	PF16451, Betacoronavirus-like spike glycoprotein S1, N-terminal
bCoV_S1_RBD	349..526(4.1e-41)	Detail	PF09408, Betacoronavirus spike glycoprotein S1, receptor binding
CoV_S1_C	536..592(1.7e-07)	Detail	PF19209, Coronavirus spike glycoprotein S1, C-terminal
CoV_S2_C	1233..1263(0.00033)	Detail	PF19214, Coronavirus spike glycoprotein S2, intravirion
Baculo_PEP_C	925..989(0.14)	Detail	PF04513, Baculovirus polyhedron envelope protein, PEP, C terminus
CaM_bind	853..870(0.093)	Detail	PF16025, Calcium-dependent calmodulin binding
DUF2959	917..1016(0.078)	Detail	PF11172, Protein of unknown function (DUF2959)
MukF_M	903..969(0.18)	Detail	PF17192, MukF middle domain
TRPM_tetra	934..962(0.55)	Detail	PF16519, Tetramerisation domain of TRPM
GP41	1149..1241(0.33)	Detail	PF00517, Retroviral envelope protein
DUF3418	918..1031(0.6)	Detail	PF11898, Domain of unknown function (DUF3418)

Motif in the sequence

Pfam ID:
CoV_S2
Description:
PF01601, Coronavirus spike glycoprotein S2
Appearance:

Position	711..1232
Alignment Query Database	SIAIPTNFTISVTTEILPVSMTKTSVDCTMYICGDSTECSNLLLQYGSFCTQLNRALTGIAVEQDKNTQEVFAQVKQIYKTPPIKDFGG--FNFSQILPDPSKpsKRSFIED nvsiptnftlsvqteyiqttspkvsvdCakyvCngnerClqlLsqygsacekiekalkssarledaevksllkilkealklaeisnfdsdlinlskllpkkns..krSaieD
i-Evalue	2.5e-201

Sequence:

MFVFLVLLPLVSSQCVNLTTRTQLPPAYTNSFTRGVYYPDKVFRSSVLHSTQDLFLPFFS
NVTWFHAIHVSGTNGTKRFDNPVLPFNDGVYFASTEKSNIIRGWIFGTTLDSKTQSLLIV
NNATNVVIKVCEFQFCNDPFLGVYYHKNNKSWMESEFRVYSSANNCTFEYVSQPFLMDLE
GKQGNFKNLREFVFKNIDGYFKIYSKHTPINLVRDLPQGFSALEPLVDLPIGINITRFQT
LLALHRSYLTPGDSSSGWTAGAAAYYVGYLQPRTFLLKYNENGTITDAVDCALDPLSETK
CTLKSFTVEKGIYQTSNFRVQPTESIVRFPNITNLCPFGEVFNATRFASVYAWNRKRISN
CVADYSVLYNSASFSTFKCYGVSPTKLNDLCFTNVYADSFVIRGDEVRQIAPGQTGKIAD
YNYKLPDDFTGCVIAWNSNNLDSKVGGNYNYLYRLFRKSNLKPFERDISTEIYQAGSTPC
NGVEGFNCYFPLQSYGFQPTNGVGYQPYRVVVLSFELLHAPATVCGPKKSTNLVKNKCVN
FNFNGLTGTGVLTESNKKFLPFQQFGRDIADTTDAVRDPQTLEILDITPCSFGGVSVITP
GTNTSNQVAVLYQDVNCTEVPVAIHADQLTPTWRVYSTGSNVFQTRAGCLIGAEHVNNSY
ECDIPIGAGICASYQTQTNSPRRARSVASQSIIAYTMSLGAENSVAYSNNSIAIPTNFTI
SVTTEILPVSMTKTSVDCTMYICGDSTECSNLLLQYGSFCTQLNRALTGIAVEQDKNTQE
VFAQVKQIYKTPPIKDFGGFNFSQILPDPSKPSKRSFIEDLLFNKVTLADAGFIKQYGDC
LGDIAARDLICAQKFNGLTVLPPLLTDEMIAQYTSALLAGTITSGWTFGAGAALQIPFAM
QMAYRFNGIGVTQNVLYENQKLIANQFNSAIGKIQDSLSSTASALGKLQDVVNQNAQALN
TLVKQLSSNFGAISSVLNDILSRLDKVEAEVQIDRLITGRLQSLQTYVTQQLIRAAEIRA
SANLAATKMSECVLGQSKRVDFCGKGYHLMSFPQSAPHGVVFLHVTYVPAQEKNFTTAPA
ICHDGKAHFPREGVFVSNGTHWFVTQRNFYEPQIITTDNTFVSGNCDVVIGIVNNTVYDP
LQPELDSFKEELDKYFKNHTSPDVDLGDISGINASVVNIQKEIDRLNEVAKNLNESLIDL
QELGKYEQYIKWPWYIWLGFIAGLIAIVMVTIMLCCMTSCCSCLKGCCSCGSCCKFDEDD
SEPVLKGVKLHYT

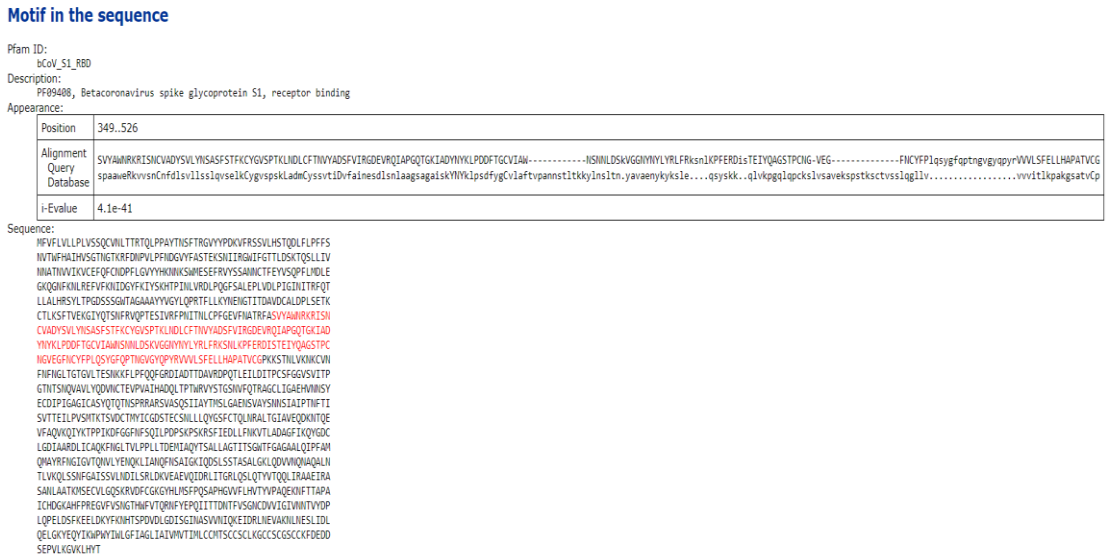
Motif in the sequence

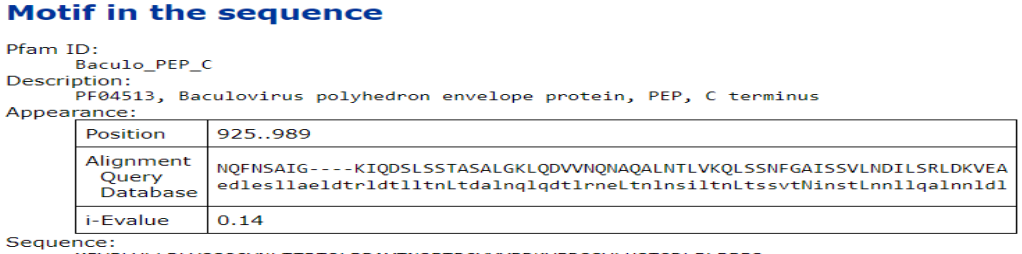
Pfam ID:
bCoV_S1_N
Description:
PF16451, Betacoronavirus-like spike glycoprotein S1, N-terminal
Appearance:

Position	33..337
Alignment Query Database	TRGVYY-PDKVFrSSVLHSTQDLFLPFFSNVTWFHAIHVSGTNGTKRFDNPVLPFNDGVYFASTEKSNIIRG----WIFGTTLDSKTQSLLIVNNA--TNVVIKVCEFQFCNDPFLGVYY--- aegvlylpdrvf.snattllqglFpkqgsnvtnylisnakskkekrlfstsvipFgdGifvkitnaknateseppavvfGstfenksatliiapdstqtnltvsvCnytlcanPvlasrkglg
i-Evalue	6.8e-70

Sequence:

MFVFLVLLPLVSSQCVNLTTRTQLPPAYTNSFTRGVYYPDKVFRSSVLHSTQDLFLPFFS
NVTWFHAIHVSGTNGTKRFDNPVLPFNDGVYFASTEKSNIIRGWIFGTTLDSKTQSLLIV
NNATNVVIKVCEFQFCNDPFLGVYYHKNNKSWMESEFRVYSSANNCTFEYVSQPFLMDLE
GKQGNFKNLREFVFKNIDGYFKIYSKHTPINLVRDLPQGFSALEPLVDLPIGINITRFQT
LLALHRSYLTPGDSSSGWTAGAAAYYVGYLQPRTFLLKYNENGTITDAVDCALDPLSETK
CTLKSFTVEKGIYQTSNFRVQPTESIVRFPNITNLCPFGEVFNATRFASVYAWNRKRISN
CVADYSVLYNSASFSTFKCYGVSPTKLNDLCFTNVYADSFVIRGDEVRQIAPGQTGKIAD
YNYKLPDDFTGCVIAWNSNNLDSKVGGNYNYLYRLFRKSNLKPFERDISTEIYQAGSTPC
NGVEGFNCYFPLQSYGFQPTNGVGYQPYRVVVLSFELLHAPATVCGPKKSTNLVKNKCVN
FNFNGLTGTGVLTESNKKFLPFQQFGRDIADTTDAVRDPQTLEILDITPCSFGGVSVITP
GTNTSNQVAVLYQDVNCTEVPVAIHADQLTPTWRVYSTGSNVFQTRAGCLIGAEHVNNSY
ECDIPIGAGICASYQTQTNSPRRARSVASQSIIAYTMSLGAENSVAYSNNSIAIPTNFTI
SVTTEILPVSMTKTSVDCTMYICGDSTECSNLLLQYGSFCTQLNRALTGIAVEQDKNTQE
VFAQVKQIYKTPPIKDFGGFNFSQILPDPSKPSKRSFIEDLLFNKVTLADAGFIKQYGDC
LGDIAARDLICAQKFNGLTVLPPLLTDEMIAQYTSALLAGTITSGWTFGAGAALQIPFAM
QMAYRFNGIGVTQNVLYENQKLIANQFNSAIGKIQDSLSSTASALGKLQDVVNQNAQALN
TLVKQLSSNFGAISSVLNDILSRLDKVEAEVQIDRLITGRLQSLQTYVTQQLIRAAEIRA
SANLAATKMSECVLGQSKRVDFCGKGYHLMSFPQSAPHGVVFLHVTYVPAQEKNFTTAPA
ICHDGKAHFPREGVFVSNGTHWFVTQRNFYEPQIITTDNTFVSGNCDVVIGIVNNTVYDP
LQPELDSFKEELDKYFKNHTSPDVDLGDISGINASVVNIQKEIDRLNEVAKNLNESLIDL
QELGKYEQYIKWPWYIWLGFIAGLIAIVMVTIMLCCMTSCCSCLKGCCSCGSCCKFDEDD
SEPVLKGVKLHYT





- **Mutation Analysis using Uniprot Variant Viewer and NCBI Family Variation: -**

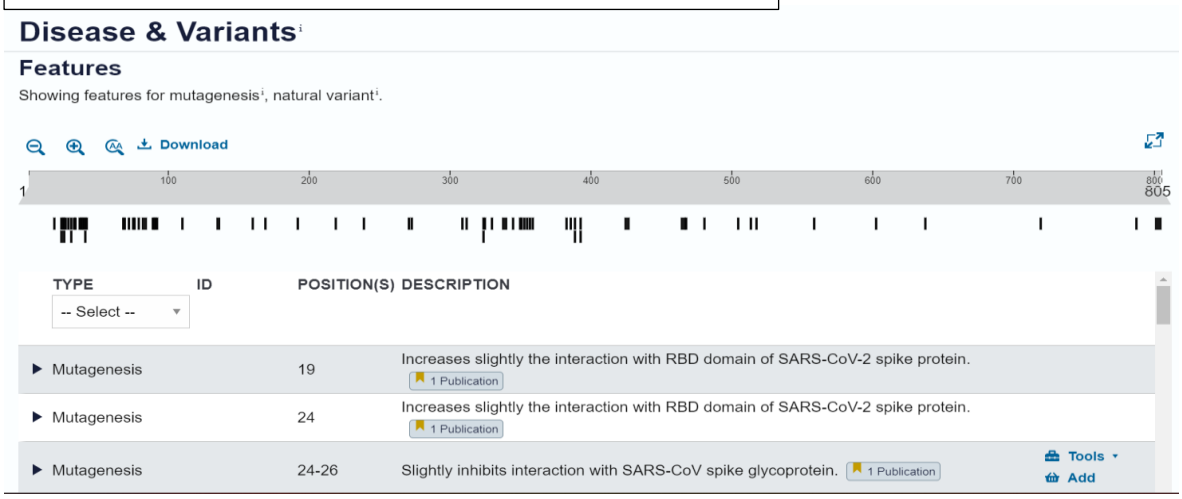
The SARS-CoV-2 spike protein and the ACE2 receptor will give evidence through mutation analysis of differences that might impact viral infection and binding. An investigator might use UniProt Variant Viewer and NCBI BLAST Distant Family Tree to assist in finding known mutations, the clinical impact of these mutations, and evolutionary effects. It will further help in establishing a possible influence of variants on drug efficacy and vaccine advancement.

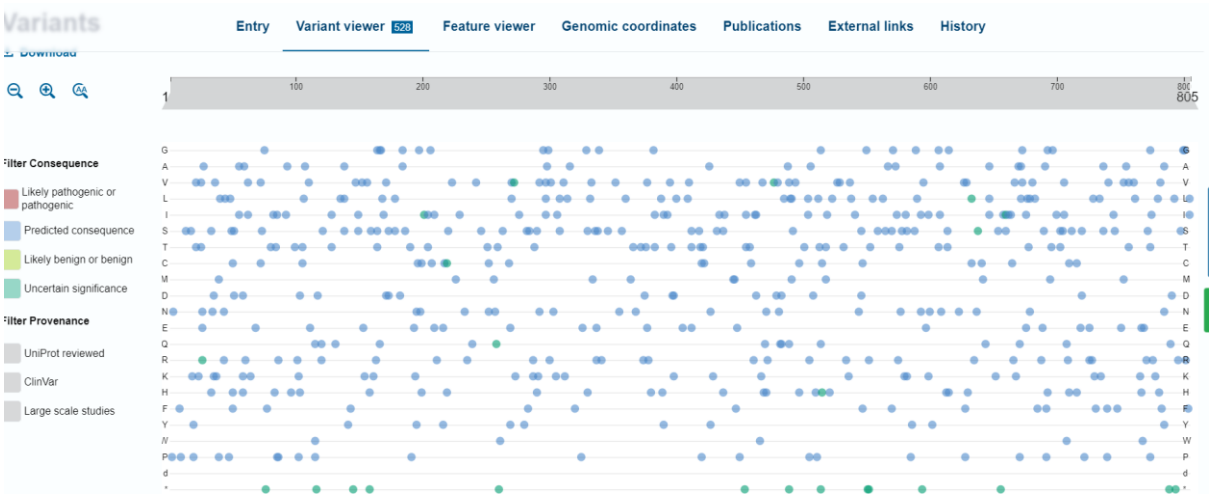
- **<u>Steps for Mutation Analysis:</u>**

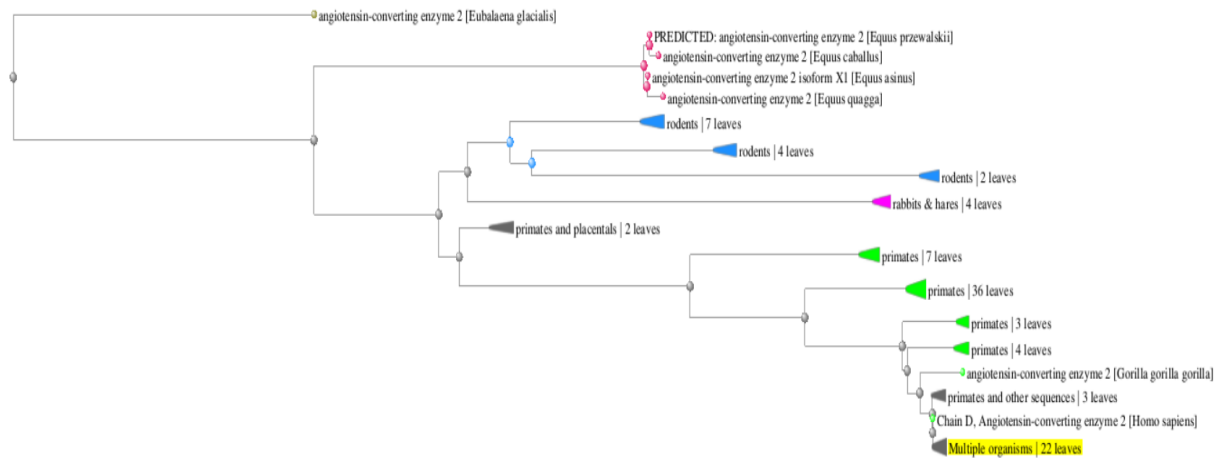
1. **Access UniProt Variant Viewer**:
    - Go to the UniProt website and search for ACE2 or spike protein entries.

2. **Review Variants**:
    - Navigate to the "Variants" section to view documented mutations and their effects.
3. **Access NCBI BLAST Distatnt Family Result**:
    - Visit the NCBI BLAST Family Result and search for ACE2 or spike protein

Mutation and variant analysis result for human ace2 protein





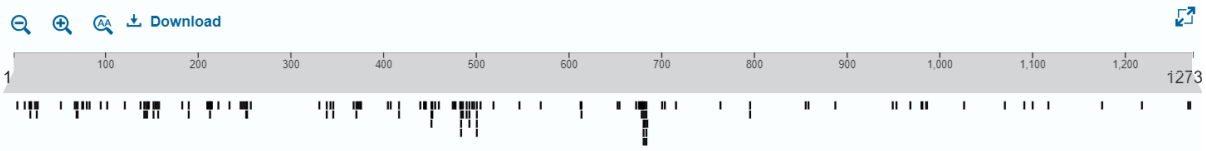


Mutation and variant analysis result for sars-cov2 spike glycoprotein

Phenotypes & Variants

Features

Showing features for natural variant, mutagenesis.



TYPE	ID	POSITION(S)	DESCRIPTION
-- Select --			
Natural variant		5	in strain: Iota/B.1.526 Curated
Natural variant		13	in strain: Epsilon/B.1.427/B.1.429; may change signal peptide cleavage to position 16-17 1 Publication Curated
Natural variant		18	in strain: Beta/B.1.351, Gamma/P.1, 19B/501Y 2 Publications
Natural variant		19	in strain: Omicron/BQ.1.1, Omicron/XBB.1.5, Omicron/EG.5.1 Curated





(v). Antibody and Epitope Prediction

- IEDB B Cell Tools and T-cell Tools:

Run Spike and ACE2 sequences using IEDB tools to predict B-cell and T-cell epitopes in studying immune response; this will focus on regions important for viral entry and immune evasion.

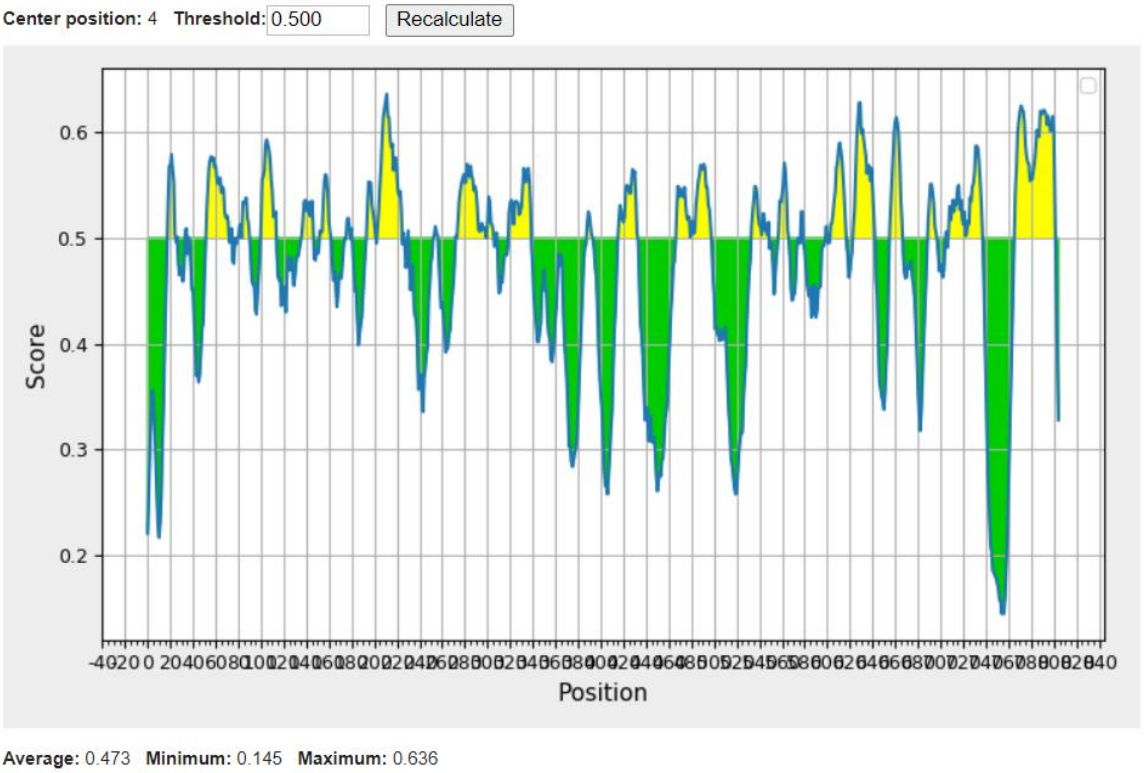
- ***IEDB B Cell Tools (Epitope Prediction)***

Use *Linear Epitope* and *Discotope* tools for the prediction of B-cell epitopes. Spike and ACE2 sequences shall be run against these tools to find out surface-exposed regions of importance for immune recognition and binding.

- ***IEDB T-cell Tools (MHC Class I and II Prediction)***

Tools utilized for T-cell epitope prediction include **MHC Class I** and **MHC Class II**. The sequences of Spike and ACE2 were uploaded to predict peptide binding to the MHC molecules, keeping the antigen presentation and immune response in view.

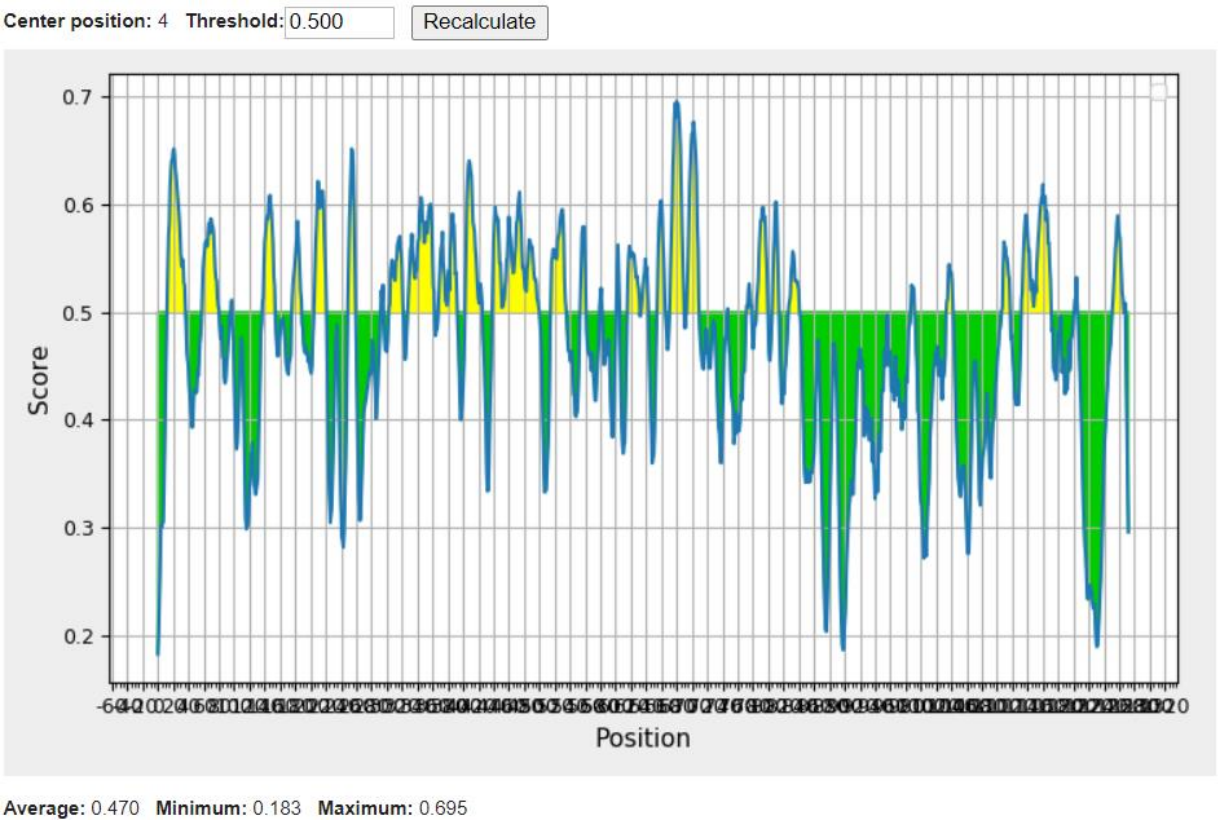
IEDB B Cell Tool Bepipred Linear Epitope Prediction Data
Result of Human ACE2 protein



The graph represents scores at different positions, likely related to some form of testing or quality control. Here are the key takeaways:

- **Threshold**: The baseline threshold is set at 0.500.
- **Score Fluctuations**: The yellow line fluctuates above and below the threshold.
- **Significantly Below Threshold**: The green areas indicate scores well below the threshold.
- **Summary Statistics**: The average score is 0.473, with a minimum of 0.145 and a maximum of 0.636.

IEDB B Cell Tool Bepipred Linear Epitope Prediction Data
Result of SARS CoV-2 Spike Glycoprotein

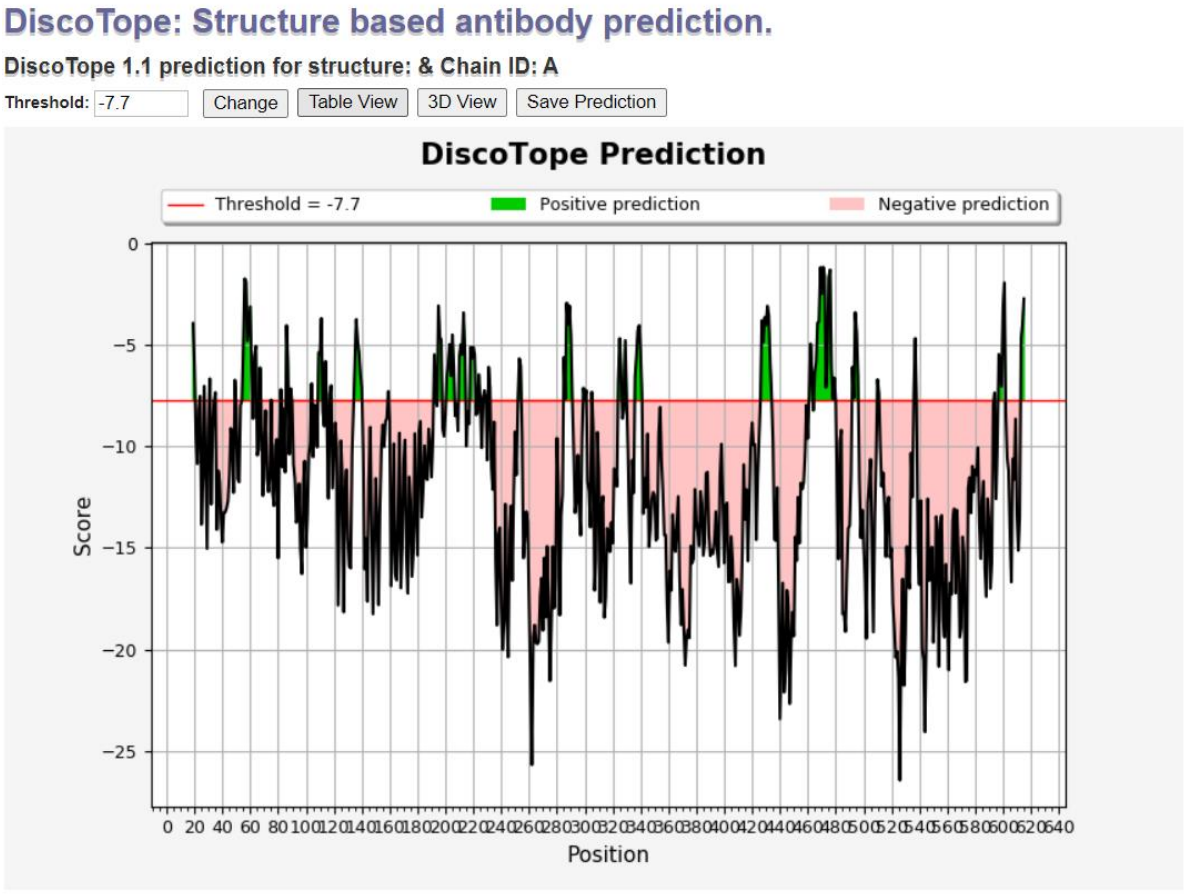


The graph represents scores at different positions, likely related to some form of testing or quality control. Here are the key takeaways:

- **Threshold**: The baseline threshold is set at 0.500.
- **Score Fluctuations**: The yellow line fluctuates above and below the threshold.
- **Significantly Below Threshold**: The green areas indicate scores well below the threshold.

- **Summary Statistics**: The average score is 0.470, with a minimum of 0.183 and a maximum of 0.695

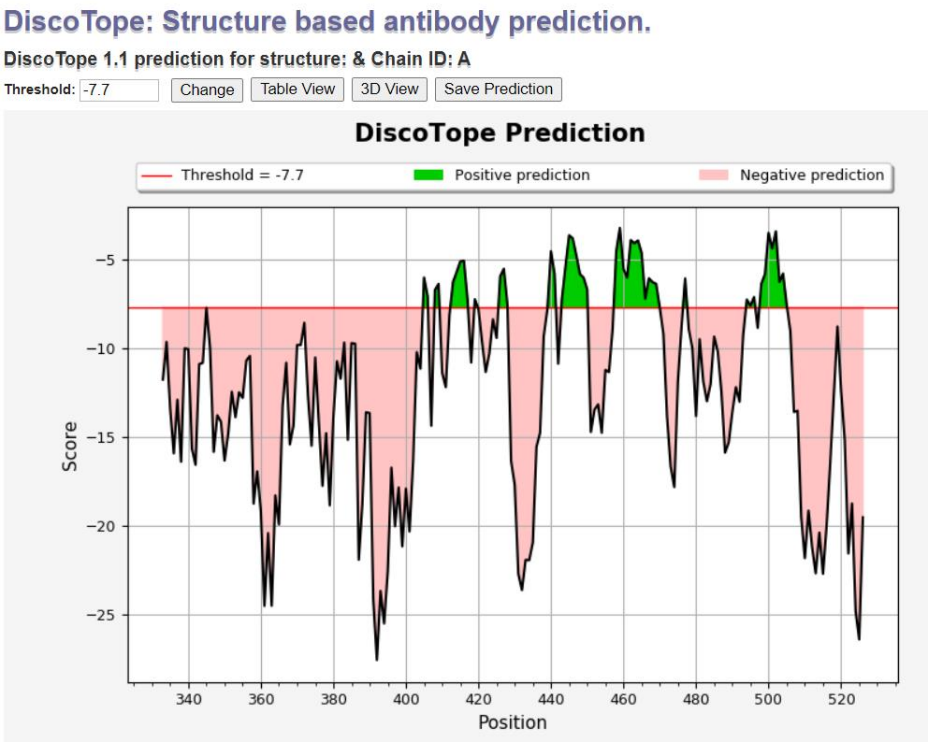
IEDB B Cell Tool DiscoTope: Structure based antibody prediction Data Result of Human ACE2 Protein



The graph represents scores at different positions, likely related to some form of testing or quality control. Here are the key takeaways:

- **Threshold**: The baseline threshold is set at 0.500.
- **Score Fluctuations**: The yellow line fluctuates above and below the threshold.
- **Significantly Below Threshold**: The green areas indicate scores well below the threshold.
- **Summary Statistics**: The average score is 0.470, with a minimum of 0.183 and a maximum of 0.695

IEDB B Cell Tool DiscoTope: Structure based antibody prediction Data Result of SARS CoV-2 Spike Glycoprotein



The graph represents scores at different positions, likely related to some form of testing or quality control. Here are the key takeaways:

- **Threshold**: The baseline threshold is set at 0.500.
- **Score Fluctuations**: The yellow line fluctuates above and below the threshold.
- **Significantly Below Threshold**: The green areas indicate scores well below the threshold.
- **Summary Statistics**: The average score is 0.470, with a minimum of 0.183 and a maximum of 0.695.

IEDB T Cell Tool MHC-I Binding Prediction Data Result of Human ACE2 Protein

The result displays a table titled "MHC-I Binding Prediction Results" related to the **Human ACE2 Protein**. In this table, various alleles (such as HLA-A*01:01 and HLA-B*07:02) are listed alongside peptide sequences. Each peptide represents a sequence of amino acids from the ACE2 protein. The table also provides scores and percentile ranks, which indicate the binding affinity of these peptides to MHC-I molecules

MHC-I Binding Prediction Results

Method used: netmhcpan_el

allele	seq_num	start	end	length	peptide	score	percentile_rank	
HLA-A*01:01	1	506	515	10	VSNDYSFIRY	0.915268	0.03	
HLA-A*01:01	1	797	805	9	NTDDVQTSF	0.887097	0.04	
HLA-A*01:01	1	228	237	10	HTFEEIKPLY	0.876753	0.04	
HLA-A*01:01	1	487	497	11	VVEPVPHDETY	0.761839	0.08	
HLA-A*01:01	1	507	515	9	SNDYSFIRY	0.705187	0.1	
HLA-A*01:01	1	370	381	12	LTAHHEMGHIQY	0.691859	0.1	
HLA-A*01:01	1	42	50	9	QSSLASWNY	0.670045	0.1	
HLA-A*01:01	1	28	41	14	FLDKFNHEAEDLFY	0.629447	0.12	
HLA-A*01:01	1	796	805	10	QNTDDVQTSF	0.615871	0.13	
HLA-A*01:01	1	505	515	11	HVSNDYSFIRY	0.569927	0.15	
HLA-A*01:01	1	506	516	11	VSNDYSFIRYY	0.514306	0.18	
HLA-A*01:01	1	795	805	11	FQNTDDVQTSF	0.512472	0.18	
HLA-A*01:01	1	497	510	14	YCDPASLFHVSNDY	0.510997	0.19	
HLA-A*01:01	1	333	341	9	LTDPGNVQK	0.505339	0.19	
HLA-A*01:01	1	495	505	11	ETYCDPASLFH	0.446118	0.23	
HLA-A*01:01	1	41	50	10	YQSSLASWNY	0.433248	0.24	
HLA-A*01:01	1	169	180	12	RSEVGKQLRPLY	0.427073	0.25	
HLA-A*01:01	1	497	505	9	YCDPASLFH	0.417099	0.25	
HLA-A*01:01	1	371	381	11	TAHHEMGHIQY	0.406077	0.26	
HLA-A*01:01	1	507	516	10	SNDYSFIRYY	0.393449	0.27	
HLA-A*01:01	1	547	555	9	STEAGQKLF	0.369739	0.29	
HLA-A*01:01	1	119	127	9	ILNTMSTIY	0.361108	0.3	
HLA-A*01:01	1	641	649	9	YLFRSSVAY	0.353692	0.31	
HLA-A*01:01	1	229	237	9	TFEEIKPLY	0.349651	0.31	
HLA-A*01:01	1	446	454	9	IVGTLPFTY	0.333135	0.33	
HLA-A*01:01	1	199	207	9	YGDYWRGDY	0.319805	0.35	
HLA-A*01:01	1	32	41	10	FNHEAEDLFY	0.307406	0.36	
HLA-A*01:01	1	625	633	9	KSALGDKAY	0.283371	0.39	
HLA-A*01:01	1	75	83	9	EQSTLAQMY	0.276808	0.41	
HLA-A*01:01	1	332	341	10	MLTDPGNVQK	0.27534	0.42	
HLA-A*01:01	1	117	127	11	NTILNTMSTIY	0.264155	0.43	
HLA-A*01:01	1	54	62	9	ITEENVQNM	0.253329	0.44	
HLA-A*01:01	1	28	36	9	FLDKFNHEA	0.236777	0.48	
HLA-A*01:01	1	227	237	11	EHTFEEIKPLY	0.212733	0.51	
HLA-A*01:01	1	645	654	10	SSVAYAMRQY	0.212722	0.51	
HLA-A*01:01	1	208	217	10	EVNGVDGYDY	0.208906	0.52	
HLA-A*01:01	1	646	654	9	SVAYAMRQY	0.203224	0.54	
HLA-A*01:01	1	235	243	9	PLYEHLHAY	0.193364	0.56	
HLA-A*01:01	1	502	510	9	SLFHVSNDY	0.183651	0.58	
HLA-A*01:01	1	501	510	10	ASLFHVSNDY	0.181208	0.58	
HLA-A*01:01	1	444	454	11	LTIVGTLPFTY	0.174758	0.61	
HLA-A*01:01	1	175	183	9	QLRPLYEEY	0.174392	0.61	
HLA-A*01:01	1	365	373	9	TMDDFLTAH	0.168145	0.62	
HLA-A*01:01	1	562	570	9	KSFPWTLAL	0.164464	0.64	

IEDB T Cell Tool MHC-I Binding Prediction Data Result of SARS CoV-2 Spike Glycoprotein

The MHC-I binding prediction data for the SARS-CoV-2 spike glycoprotein reveals that different MHC-I alleles exhibit varying affinities for specific peptides. High scores indicate stronger predicted binding, which could be crucial for designing effective vaccines.

MHC-I Binding Prediction Results

Method used: netmhcpan_el

allele	seq_num	start	end	length	peptide	score	percentile_rank	
HLA-A*01:01	1	865	873	9	LTDEMIAQY	0.997196	0.01	
HLA-A*01:01	1	864	873	10	LLTDEMIAQY	0.961805	0.01	
HLA-A*01:01	1	604	612	9	TSNQVAVLY	0.917927	0.03	
HLA-A*01:01	1	746	756	11	STECSNLLLQY	0.916806	0.03	
HLA-A*01:01	1	828	837	10	LADAGFIKQY	0.840121	0.05	
HLA-A*01:01	1	258	266	9	WTAGAAAYY	0.81389	0.06	
HLA-A*01:01	1	1197	1206	10	LIDLQELGKY	0.791918	0.07	
HLA-A*01:01	1	28	38	11	YTNSFTRGVYY	0.77295	0.07	
HLA-A*01:01	1	866	873	8	TDEMIAQY	0.754765	0.08	
HLA-A*01:01	1	603	612	10	NTSNQVAVLY	0.744843	0.08	
HLA-A*01:01	1	861	873	13	LPPLLTDEMIAQY	0.730038	0.09	
HLA-A*01:01	1	863	873	11	PLLTDEMIAQY	0.724299	0.09	
HLA-A*01:01	1	652	660	9	GAEHVNNSY	0.698791	0.1	
HLA-A*01:01	1	440	449	10	NLDSKVGGNY	0.696259	0.1	
HLA-A*01:01	1	161	170	10	SSANNCTFEY	0.691415	0.1	
HLA-A*01:01	1	362	369	8	VADYSVLY	0.681243	0.1	
HLA-A*01:01	1	28	37	10	YTNSFTRGVY	0.661513	0.1	
HLA-A*01:01	1	160	170	11	YSSANNCTFEY	0.653516	0.11	
HLA-A*01:01	1	361	369	9	CVADYSVLY	0.64603	0.11	
HLA-A*01:01	1	733	741	9	KTSVDCTMY	0.642966	0.12	
HLA-A*01:01	1	196	204	9	NIDGYFKIY	0.616114	0.13	
HLA-A*01:01	1	30	38	9	NSFTRGVYY	0.597285	0.14	
HLA-A*01:01	1	860	873	14	VLPPLLTDEMIAQY	0.587166	0.14	
HLA-A*01:01	1	440	451	12	NLDSKVGGNYNY	0.576144	0.15	
HLA-A*01:01	1	1144	1155	12	ELDSFKEELDKY	0.572337	0.15	
HLA-A*01:01	1	162	170	9	SANNCTFEY	0.570386	0.15	
HLA-A*01:01	1	865	874	10	LTDEMIAQYT	0.563645	0.16	
HLA-A*01:01	1	862	873	12	PPLLTDEMIAQY	0.556798	0.16	
HLA-A*01:01	1	1260	1272	13	DSEPVLKGVKLHY	0.55165	0.17	
HLA-A*01:01	1	1039	1047	9	RVDFCGKGY	0.526344	0.18	
HLA-A*01:01	1	358	369	12	ISNCVADYSVLY	0.511532	0.19	
HLA-A*01:01	1	695	707	13	YTMSLGAENSVAY	0.508	0.19	
HLA-A*01:01	1	601	612	12	GTNTSNQVAVLY	0.502811	0.19	
HLA-A*01:01	1	256	266	11	SGWTAGAAAYY	0.492769	0.19	
HLA-A*01:01	1	651	660	10	IGAEHVNNSY	0.485311	0.2	
HLA-A*01:01	1	625	636	12	HADQLTPTWRVY	0.481939	0.2	
HLA-A*01:01	1	1146	1155	10	DSFKEELDKY	0.481064	0.2	
HLA-A*01:01	1	697	707	11	MSLGAENSVAY	0.461789	0.21	
HLA-A*01:01	1	269	279	11	YLQPRTFLLKY	0.452768	0.23	
HLA-A*01:01	1	370	380	11	NSASFSTFKCY	0.452445	0.23	
HLA-A*01:01	1	414	423	10	QTGKIADYNY	0.437978	0.24	
HLA-A*01:01	1	1197	1209	13	LIDLQELGKYEQY	0.430121	0.24	
HLA-A*01:01	1	687	695	9	VASQSIIAY	0.42647	0.25	
HLA-A*01:01	1	135	144	10	FCNDPFLGVY	0.410856	0.26	

IEDB T Cell Tool MHC-II Binding Prediction Data Result of Human ACE2 Protein

MHC-II Binding Prediction Results

Input Sequences

#	Name	Sequence
1	NP_001358344.1 angiotensin-converting enzyme 2 isoform 1 precursor [Homo sapiens]	MSSSSWLLLSLVAVTAAQSTIEEQAKTFLDKFNHEAEDLFYQSSLASWNY NTNITEENVQNMNNAGDKWSAFLKEQSTLAQMYPLQEIQNLTVKLQLQAL QQNGSSVLSEDKSKRLNTILNTMSTIYSTGKVCNPDNPQECLLLEPGLNE IMANSLDYNERLWAWESWRSEVGKQLRPLYEEYVVLKNEMARANHYEDYG DYWRGDYEVNGVDGYDYSRGQLIEDVEHTFEEIKPLYEHLHAYVRAKLMN AYPSYISPIGCLPAHLLGDMWGRFWTNLYSLTVPFGQKPNIDVTDAMVDQ AWDAQRIFKEAEKFFVSVGLPNMTQGFWENSMLTDPGNVQKAVCHPTAWD LGKGDFRILMCTKVTMDDFLTAHHEMGHIQYDMAYAAQPFLLRNGANEGF HEAVGEIMSLSAATPKHLKSIGLLSPDFQEDNETEINFLLKQALTIVGTL PFTYMLEKWRWMVFKGEIPKDQWMKKWWEMKREIVGVVEPVPHDETYCDP ASLFHVSNDYSFIRYYTRTLYQFQFQEALCQAAKHEGPLHKCDISNSTEA GQKLFNMLRLGKSEPWTLALENVVGAKNMNVRPLLNYFEPLFTWLKDQNK NSFVGWSTDWSPYADQSIKVRISLKSALGDKAYEWNDNEMYLFRSSVAYA MRQYFLKVKNQMILFGEEDVRVANLKPRISFNFFVTAPKNVSDIIPRTEV EKAIRMSRSRINDAFRLNDNSLEFLGIQPTLGPPNQPPVSIWLIVFGVVM GVIVVGIVILIFTGIRDRKKKNKARSGENPYASIDISKGENNPGFQNTDD VQTSF

Prediction method: netmhciipan_el 4.1 | High score = good binders

Download result

Citations

Allele	#	Start	End	Length	Core Sequence	Peptide Sequence	Score	Percentile Rank
HLA-DRB1*01:01	1	180	194	15	YVVLKNEMA	YEEYVVLKNEMARAN	0.9801	0.03
HLA-DRB1*01:01	1	179	193	15	YVVLKNEMA	LYEEYVVLKNEMARA	0.9708	0.08
HLA-DRB1*01:01	1	181	195	15	YVVLKNEMA	EEYVVLKNEMARANH	0.9597	0.13
HLA-DRB1*01:01	1	178	192	15	YVVLKNEMA	PLYEEYVVLKNEMAR	0.9429	0.26
HLA-DRB1*01:01	1	164	178	15	WRSEVGKQL	AWESWRSEVGKQLRP	0.9220	0.34
HLA-DRB1*01:01	1	165	179	15	WRSEVGKQL	WESWRSEVGKQLRPL	0.9095	0.41
HLA-DRB1*01:01	1	255	269	15	IGCLPAHLL	YISPIGCLPAHLLGD	0.8780	0.56
HLA-DRB1*01:01	1	212	226	15	YDYSRGQLI	VDGYDYSRGQLIEDV	0.8669	0.58
HLA-DRB1*01:01	1	256	270	15	IGCLPAHLL	ISPIGCLPAHLLGDM	0.8684	0.58
HLA-DRB1*01:01	1	353	367	15	FRILMCTKV	KGDFRILMCTKVTMD	0.8639	0.60
HLA-DRB1*01:01	1	166	180	15	WRSEVGKQL	ESWRSEVGKQLRPLY	0.8384	0.70
HLA-DRB1*01:01	1	211	225	15	YDYSRGQLI	GVDGYDYSRGQLIED	0.8413	0.70
HLA-DRB1*01:01	1	163	177	15	WRSEVGKQL	WAWESWRSEVGKQLR	0.7609	1.30
HLA-DRB1*01:01	1	352	366	15	FRILMCTKV	GKGDFRILMCTKVTM	0.7415	1.40

Exploring MHC-II binding predictions reveals how specific peptides from the human ACE2 protein interact with different alleles, crucial for understanding immune responses. Each allele serves as a unique identifier for presenting peptides. The table includes the peptide's position in ACE2, the core sequence vital for binding, a score indicating binding strength, and a percentile rank comparing binding efficacy to others. Analyzing these metrics helps identify ACE2 segments likely to activate T cells, guiding immunotherapy strategies and enhancing our grasp of host-pathogen interactions. While science isn't infallible, it remains our best tool for insights into immunology.

IEDB T Cell Tool MHC-II Binding Prediction Data Result of SARS CoV-2 Spike Glycoprotein

The table outlines predictions for peptide binding to MHC-II alleles, which are crucial for immune responses. Each allele represents a variant of the MHC-II molecule with specific binding properties. The peptide sequences are tested for their ability to bind, highlighting the core sequences essential for binding. Key metrics include the start and end positions of the peptides, their lengths, and scores reflecting binding strength—higher scores indicate better binding potential. The percentile rank shows how the binding scores compare to other peptides, with lower ranks indicating stronger binding. Examples include peptides from the SARS-CoV-2 spike protein with varying scores and ranks.

MHC-II Binding Prediction Results

Input Sequences

#	Name	Sequence
1	sp|P0DTC2.1|SPIKE_SARS2 RecName: Full=Spike glycoprotein; Short=S glycoprotein; AltName: Full=E2; AltName: Full=Peplomer protein; Contains: RecName: Full=Spike protein S1; Contains: RecName: Full=Spike protein S2; Contains: RecName: Full=Spike protein S2'; Flags: Precursor	MFVFLVLLPLVSSQCVNLTTRTQLPPAYTNSFTRGVYYPDKVFRSSVLHS TQDLFLPFFSNVTWFHAIHVSGTNGTKRFDNPVLPFNDGVYFASTEKSNI IRGWIFGTTLDSKTQSLLIVNNATNVVIKVCEFQFCNDPFLGVYYHKNNK SWMESEFRVYSSANNCTFEYVSQPFLMDLEGKQGNFKNLREFVFKNIDGY FKIYSKHTPINLVRDLPQGFSALEPLVDLPIGINITRFQTLLALHRSYLT PGDSSSGWTAGAAAYYVGYLQPRTFLLKYNENGTITDAVDCALDPLSETK CTLKSFTVEKGIYQTSNFRVQPTESIVRFPNITNLCPFGEVFNATRFASV YAWNRKRISNCVADYSVLYNSASFSTFKCYGVSPTKLNDLCFTNVYADSF VIRGDEVRQIAPGQTGKIADYNYKLPDDFTGCVIAWNSNNLDSKVGGNYN YLYRLFRKSNLKPFERDISTEIYQAGSTPCNGVEGFNCYFPLQSYGFQPT NGVGYQPYRVVVLSFELLHAPATVCGPKKSTNLVKNKCVNFNFNGLTGTG VLTESNKKFLPFQQFGRDIADTTDAVRDPQTLEILDITPCSFGGVSVITP GTNTSNQVAVLYQDVNCTEVPVAIHADQLTPTWRVYSTGSNVFQTRAGCL IGAEHVNNSYECDIPIGAGICASYQTQTNSPRRARSVASQSIIAYTMSLG AENSVAYSNNSIAIPTNFTISVTTEILPVSMTKTSVDCTMYICGDSTECS NLLLQYGSFCTQLNRALTGIAVEQDKNTQEVFAQVKQIYKTPPIKDFGGF NFSQILPDPSKPSKRSFIEDLLFNKVTLADAGFIKQYGDCLGDIAARDLI CAQKFNGLTVLPPLLTDEMIAQYTSALLAGTITSGWTFGAGAALQIPFAM QMAYRFNGIGVTQNVLYENQKLIANQFNSAIGKIQDSLSSTASALGKLQD VVNQNAQALNTLVKQLSSNFGAISSVLNDILSRLDKVEAEVQIDRLITGR LQSLQTYVTQQLIRAAEIRASANLAATKMSECVLGQSKRVDFCGKGYHLM SFPQSAPHGVVFLHVTYVPAQEKNFTTAPAICHDGKAHFPREGVFVSNGT HWFVTQRNFYEPQIITTDNTFVSGNCDVVIGIVNNTVYDPLQPELDSFKE ELDKYFKNHTSPDVDLGDISGINASVVNIQKEIDRLNEVAKNLNESLIDL QELGKYEQYIKWPWYIWLGFIAGLIAIVMVTIMLCCMTSCCSCLKGCCSC GSCCKFDEDDSEPVLKGVKLHYT

Prediction method: netmhciipan_el 4.1 | High score = good binders

Download result

Citations

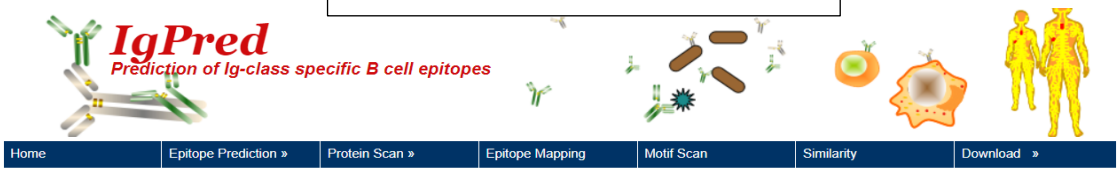
Allele	#	Start	End	Length	Core Sequence	Peptide Sequence	Score	Percentile Rank
HLA-DRB1*01:01	1	540	554	15	FNGLTGTGV	NFNFNGLTGTGVLTE	0.9272	0.32
HLA-DRB1*01:01	1	315	329	15	FRVQPTESI	TSNFRVQPTESIVRF	0.8972	0.50
HLA-DRB1*01:01	1	314	328	15	FRVQPTESI	QTSNFRVQPTESIVR	0.8941	0.51
HLA-DRB1*01:01	1	539	553	15	FNGLTGTGV	VNFNFNGLTGTGVLT	0.8606	0.61
HLA-DRB1*01:01	1	512	526	15	FELLHAPAT	VLSFELLHAPATVCG	0.8447	0.67
HLA-DRB1*01:01	1	510	524	15	FELLHAPAT	VVVLSFELLHAPATV	0.8119	0.86
HLA-DRB1*01:01	1	1057	1071	15	VVFLHVTYV	PHGVVFLHVTYVPAQ	0.7616	1.30
HLA-DRB1*01:01	1	511	525	15	FELLHAPAT	VVLSFELLHAPATVC	0.7428	1.40
HLA-DRB1*01:01	1	303	317	15	FTVEKGIYQ	LKSFTVEKGIYQTSN	0.7344	1.50

- ***IgPred Analysis(Protein Scan):*** For prediction using IgPred, I followed these steps:

- 1. Input sequence - the amino acid sequence of the target protein to predict B-cell epitopes.
- 2. I chose an appropriate prediction model considering the sequence type and the level of specificity required ; I used protein scan.
- 3. The analysis was run; and a list with the predicted epitopes and their scores was created.

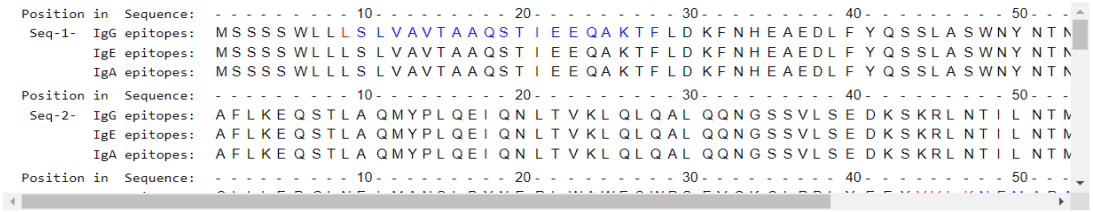
Result Analysis. The output contained regions of the sequence that are likely to elicit an immune response.There were some regions that stood out and appear to have high-scoring epitopes, which may present as such potential as a vaccine or therapeutic target for SARS-CoV-2. Such information is useful to understand the immune landscape that may serve to further experimentative validation. It gave me regions of IgE, IgA, IgA epitopes.

IgPred Result for Human ACE2 Protein

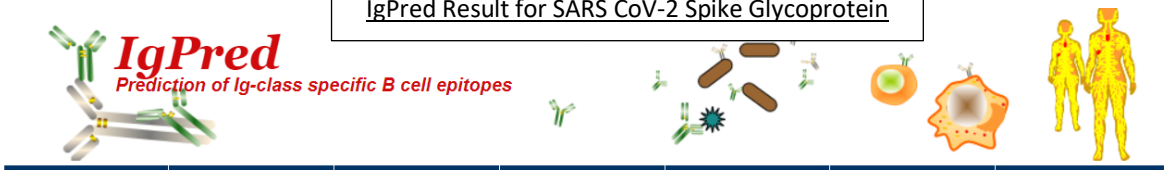


Protein scaning results:

This presentation allows user to identify and compare different antibody-class specific regions in query antigen. Epitopes are colored in blue with a starting of red color (Click here for details with score)

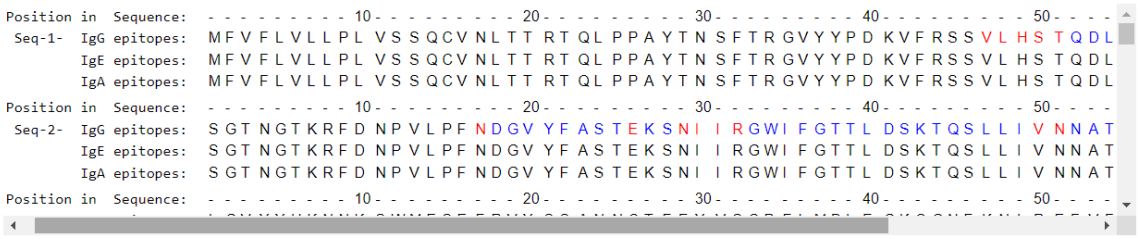


IgPred Result for SARS CoV-2 Spike Glycoprotein



Protein scaning results:

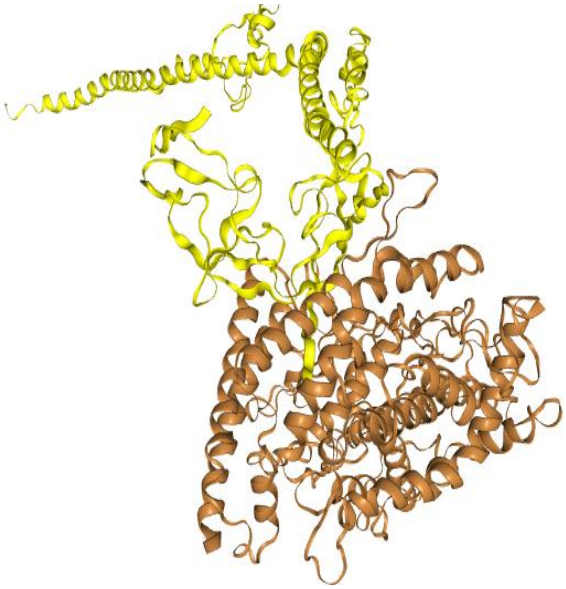
This presentation allows user to identify and compare different antibody-class specific regions in query antigen. Epitopes are colored in blue with a starting of red color (Click here for details with score)

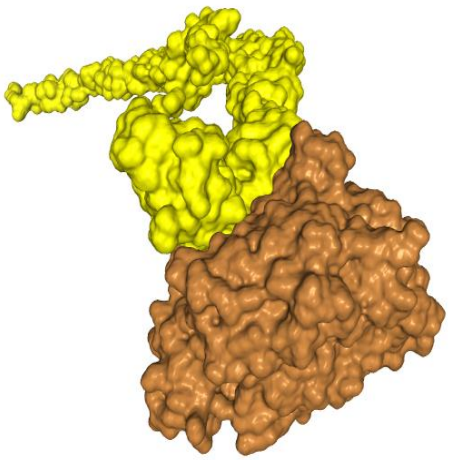


HDOCK PROTEIN-PROTEIN DOCKING RESULT

I used the two sequences of ace2 and s protein to dock them in **HDOCK Server** and scores are given below. The docking result between the human ACE2 receptor and SARS-CoV-2 spike glycoprotein using the HDOCK Server reveals very important information. The docking score of **-251.95** should be interpreted as a higher binding affinity between two proteins, since more negative values often point towards tighter interactions. The confidence score of **0.8848**suggests that the docking model is rather reliable, meaning that the higher the confidence scores, the better is the accuracy of the complex predicted. However, the ligand RMSD at **178.46 Å** is significantly large, which usually means significant deviation between the predicted and actual binding poses, so that it leads to some uncertainty about the position of the ligand.

These homology modeling quality reports, based on sequence identity and similarity (Seq_ID and TMscore), assure that both receptor and ligand structures ACE2 and the spike protein, respectively are modeled at high confidence, hence suggesting the predicted binding mode is biologically plausible. The whole results suggest a strong potential interaction between the ACE2 protein and the spike glycoprotein.





Docking Score	-251.95
Confidence Score	0.8848
Ligand rmsd (Å)	178.46

Quality of Docking Structures/Input Data

Receptor: The **protein** structure was modeled by the server.

```
=======================================================
     Quality report of homology model by TMalign
=======================================================
Template:  7DX8_D.pdb
Seq ID:    100%
TMscore:   0.99791

Different ranges of protein homology quality by Seq_ID or TMscore:
Low                      Medium                        High
Seq_ID  < 30%            30% ≤ Seq_ID  < 50%           50% ≤ Seq_ID
TMscore < 0.5            0.5 ≤ TMscore < 0.8           0.8 ≤ TMscore
```

Ligand: The **protein** structure was modeled by the server.

```
=======================================================
     Quality report of homology model by TMalign
=======================================================
Template:  6XRA_A.pdb
Seq ID:    91.9%
TMscore:   0.59647

Different ranges of protein homology quality by Seq_ID or TMscore:
Low                      Medium                        High
Seq_ID  < 30%            30% ≤ Seq_ID  < 50%           50% ≤ Seq_ID
TMscore < 0.5            0.5 ≤ TMscore < 0.8           0.8 ≤ TMscore
```

```
====== Model 1 ======

# Receptor interface residue(s):
TYR      50A      2.485
ASN      51A      4.023
ILE      54A      2.569
GLU      56A      2.375
GLN      60A      4.892
MET      62A      4.614
ASN      63A      3.889
SER     105A      3.659
LEU     116A      3.978
THR     118A      4.804
LEU     120A      3.317
ASN     121A      2.762
THR     122A      4.248
SER     124A      4.072
THR     125A      3.006
SER     128A      2.821
THR     129A      3.863
LYS     131A      1.664
GLN     139A      4.397
GLU     140A      4.506
CYS     141A      3.435
LEU     142A      3.897
LEU     143A      2.205
LEU     144A      4.834
GLU     145A      3.930
PRO     146A      3.220
LYS     187A      2.697
MET     190A      4.274
TYR     199A      1.497
TYR     202A      4.018
TRP     203A      3.401
VAL     339A      4.346
LYS     341A      1.444
ALA     342A      4.485
VAL     343A      2.718
GLU     398A      4.396
ASN     508A      4.174
ASP     509A      2.624
TYR     510A      2.984
SER     511A      4.673
ARG     514A      3.592

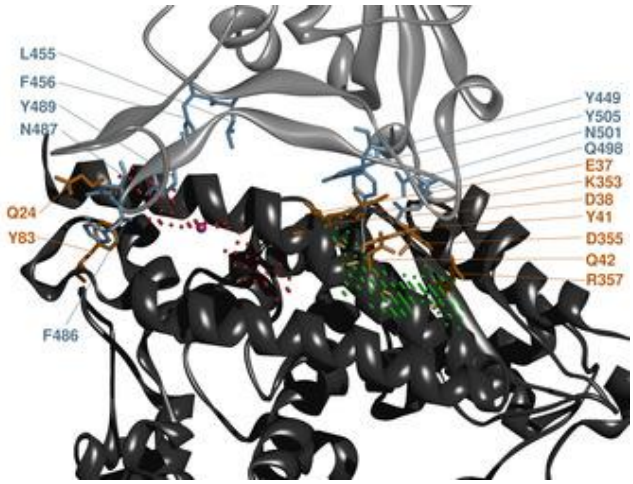
# L1gand interface residue(s):
THR      716A      2.914
ASN      717A      2.762
PHE      718A      3.006
THR      719A      4.186
ILE      720A      2.821
VAL      722A      3.220
ASN      777A      3.618
THR      778A      1.444
GLN      779A      3.401
GLU      780A      2.569
VAL      781A      3.012
PHE      782A      2.485
GLN      784A      3.540
VAL      785A      3.361
LYS      786A      2.984
GLN      787A      2.624
ILE      788A      1.497
VAL     1033A      4.622
LEU     1034A      4.640
GLN     1036A      4.941
LYS     1045A      2.375
LEU     1049A      1.664
MET     1050A      3.181
PHE     1052A      3.878
VAL     1065A      2.205
THR     1066A      4.823
TYR     1067A      3.874
PRO     1069A      4.558
GLN     1071A      4.943
GLU     1072A      3.889
ASN     1074A      4.892
PHE     1109A      3.222
TYR     1110A      4.694
GLU     1111A      4.248

# Receptor-l1gand interface residue pair(s):
50A   -   781A      4.210
50A   -   782A      2.485
50A   -  1069A      4.558
51A   -   782A      4.023
54A   -   780A      2.569
54A   -   781A      4.999
54A   -   782A      4.674
56A   -   780A      4.094
56A   -  1045A      2.375
60A   -  1074A      4.892
62A   -   782A      4.614
63A   -  1072A      3.889
105A  -   787A      3.659
116A  -   787A      3.978
118A  -  1110A      4.804
120A  -   717A      4.578
120A  -   787A      3.317
121A  -   716A      2.914
121A  -   717A      2.762
121A  -  1071A      4.943
121A  -  1109A      4.402
121A  -  1110A      4.694
122A  -  1109A      4.834
122A  -  1111A      4.248
124A  -   717A      4.072
124A  -   785A      4.236
125A  -   716A      4.935
125A  -   717A      4.808
125A  -   718A      3.006
125A  -  1109A      3.222
128A  -   718A      4.273
128A  -   719A      4.186
128A  -   720A      2.821
128A  -  1049A      4.386
129A  -   718A      4.503
129A  -  1049A      3.863
129A  -  1067A      3.874
131A  -  1036A      4.941
131A  -  1049A      1.664
131A  -  1050A      3.181
139A  -  1033A      4.622
139A  -  1034A      4.640
139A  -  1050A      4.397
140A  -  1050A      4.506
141A  -  1050A      3.435
142A  -  1052A      3.897
143A  -   720A      3.483
143A  -  1049A      3.238
143A  -  1050A      3.557
143A  -  1065A      2.205
143A  -  1066A      4.823
144A  -   720A      4.834
145A  -   720A      3.930
146A  -   720A      4.338
146A  -   722A      3.220
146A  -  1052A      3.878
146A  -  1065A      3.548
187A  -   787A      3.258
187A  -   788A      2.697
190A  -   787A      4.689
190A  -   788A      4.274
199A  -   788A      1.497
202A  -   788A      4.018
203A  -   788A      3.401
339A  -   777A      4.346
```

CONCLUSION

Now, to the whole investigation of how the SARS-CoV-2 Spike glycoprotein interacts with the human ACE2 receptor and the underlying molecular mechanisms of viral penetration and infection. The use of an array of bioinformatics tools in the study showed the detailed structural and functional dynamism that underlies regulation of the interaction. Key conserved domains, motifs, and putative binding sites were identified in the entire Spike protein and ACE2 receptor sequence using bioinformatics research, with a focus on those critical for virus attachment and invasion of host cells.

The outcomes have special significance in the design of therapeutic approaches. This allows targeted interventions in the specific regions involved, such as vaccines and antiviral drugs, which would be able to effectively block this interaction. Such therapeutics may inhibit the entry of the virus into human ACE2 and are, therefore, able to curb infection and transmission. Results from this study are not only adding to our knowledge about the pathogenicity of SARS-CoV-2 but also serving as a basis for coming up with ways to counterattack other similar viral threats. That structural and functional analysis would persist only underlines the importance of these bioinformatics approaches for the development of an effective treatment against COVID-19 and other coronaviruses.



(Cite: Pirolli D, Righino B, De Rosa MC. Targeting SARS-CoV-2 Spike Protein/ACE2 Protein-Protein Interactions: a Computational Study. Mol Inform. 2021 Jun;40(6):e2060080. doi: 10.1002/minf.202060080. Epub 2021 Apr 27. PMID: 33904240; PMCID: PMC8206717.)

ACKNOWLEDMENT

I am extremely indebted to **Dr. Ronojoy Sengupta**, Director of Genobiotek, the best bioinformatics training institution. Dr. Sengupta guided me at every stage of my research to ensure the success of this study termed "Exploring the Interaction of SARS-CoV-2 Spike Glycoprotein with Human ACE2: An Overview." Under his guidance, my knowledge pertaining to biological tools and techniques were deepened enough to successfully complete this present study.

I extend my thanks to **Dr Trinath Chowdhury**, Project Manager/Technical Head of Synthetic Moulders Limited, for encouragement and insightful suggestions that raised the quality of my research at various levels of its development, with their experience underpinnings firm support vested in making my knowledge and skill base in this field.

I will never forget their directions because it was through them that I would not have been able to carry out this research.

Genobiotek
Biotech Research and Training Laboratory



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Director, Genobiotek

M.Sc.(C.U.), Ph.D.(C.U.), PostDoc (CNU and GIST, Republic of South Korea)
Professor and Head of the Department of Biotechnology and Bioinformatics (SMMTR); Co-investigator in Nano-Biotechnology project, Variable Energy Cyclotron Center (VECC), Kolkata; Co-investigator in Reproductive Biomedicine



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Jadavpur University

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